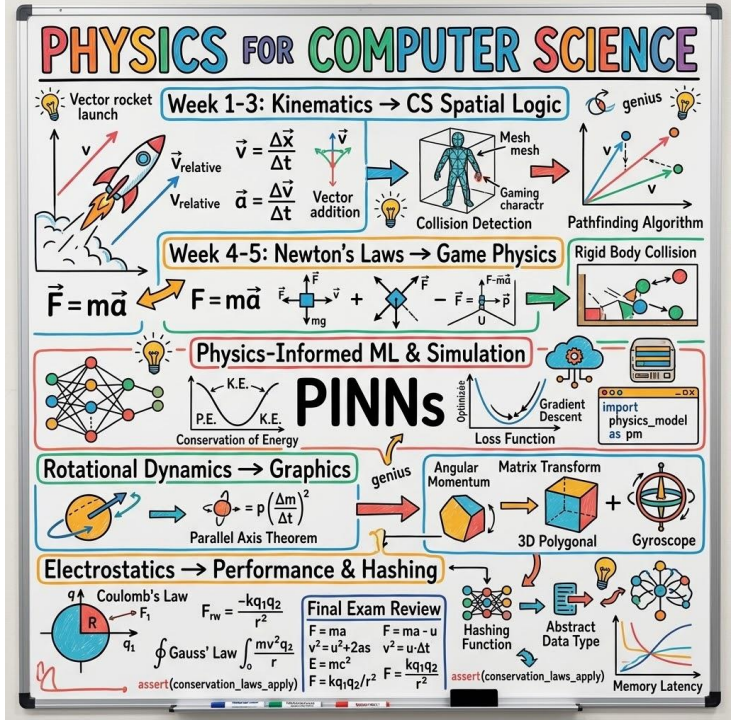
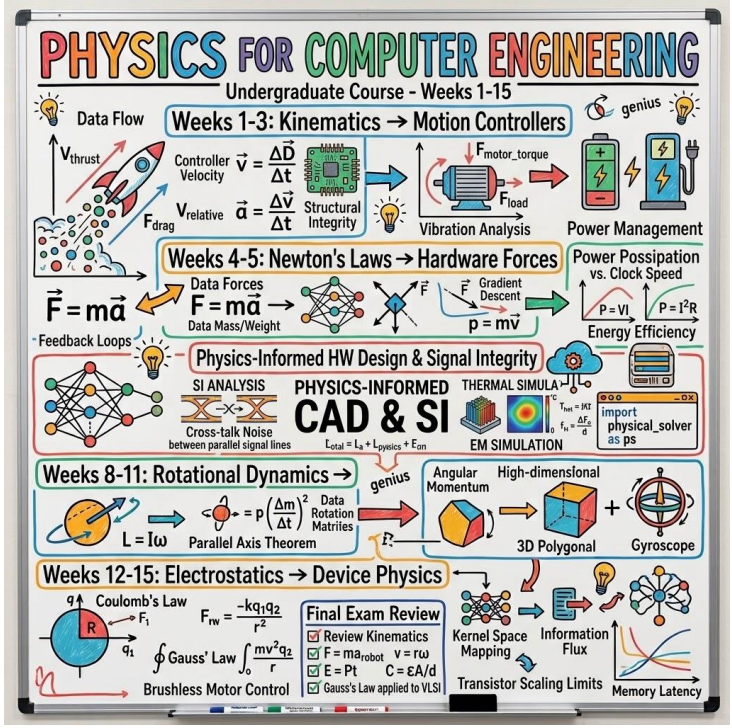
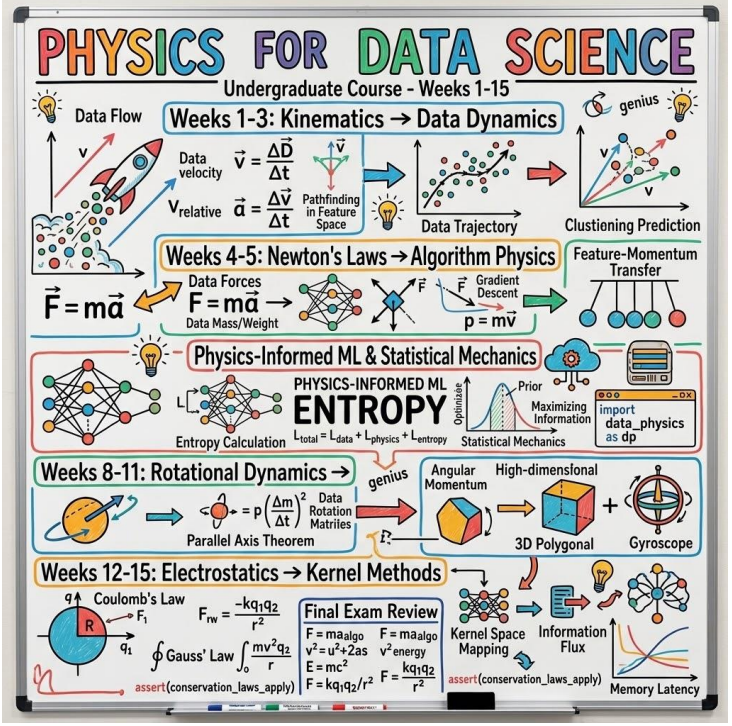
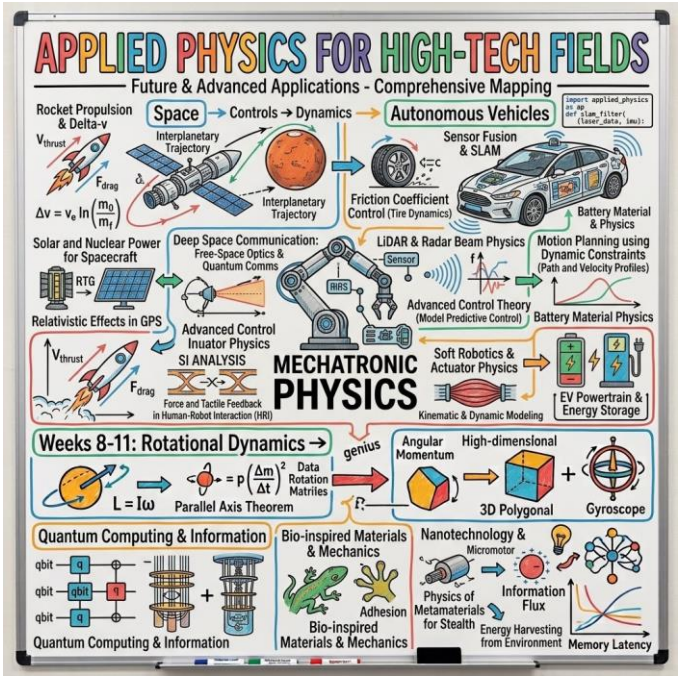








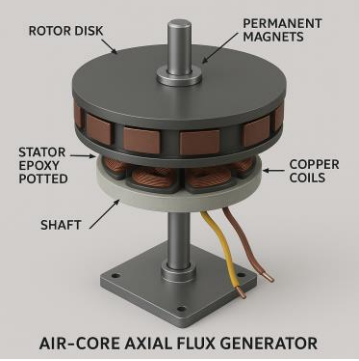
Applied Physics
PH101

Faculty of Basic Sciences

Week_05

Fall_2026

Weekly Plan



- **Newton’s Law of Motion**
- **Applying Newton’s Law**

Normalizations of vectors

Normalization condition $|\hat{A}| = 1$:

$$|\hat{A}|^2 = \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$\boxed{\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1}$$

Combination	α	β	γ	Check
Original (invalid)	$45°$	$30°$	$60°$	❌
Valid 1	$45°$	$60°$	$60°$	✅
Valid 2	$45°$	$120°$	$60°$	✅
Valid 3	$90°$	$30°$	$60°$	✅
Valid 4	$45°$	$60°$	$120°$	✅
Valid 5	$60°$	$45°$	$60°$	✅
Valid 6	$60°$	$45°$	$120°$	✅

Mechanics

Type of Mechanics	Key Focus	Sub-Types	Real-World Example
Classical	Rigid Bodies	Statics, Dynamics	Forces in a truss bridge
Continuum	Deformable Bodies	Solid Mechanics, Fluid Mechanics	Airflow over a car (aerodynamics)
Relativistic	High-Speed (≈ c)	Special & General Relativity	GPS satellite corrections
Quantum	Atomic Particles	Quantum Theory, Field Theory	Behavior of electrons in a semiconductor

Newton's 1st Laws

An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.

Inertia - the tendency of objects to resist changes in their state of motion.

$$\sum \bar{F} = 0$$

A hockey puck sliding on frictionless ice continues moving indefinitely until it hits the boards.

Newton's 2nd Laws

The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass.

Net force determines how much an object accelerates.

$$\sum \bar{F} = m\bar{a}$$

ushing a empty shopping cart (small mass) vs. a full shopping cart (large mass) with the same force - the empty cart accelerates more.

Newton's 3rd Laws

For every action, there is an equal and opposite reaction.

Forces always occur in pairs between two interacting objects.

$$\bar{F}_{12} = -\bar{F}_{21}$$

When you push against a wall, the wall pushes back on you with equal force.

Newton’s Laws of Motion

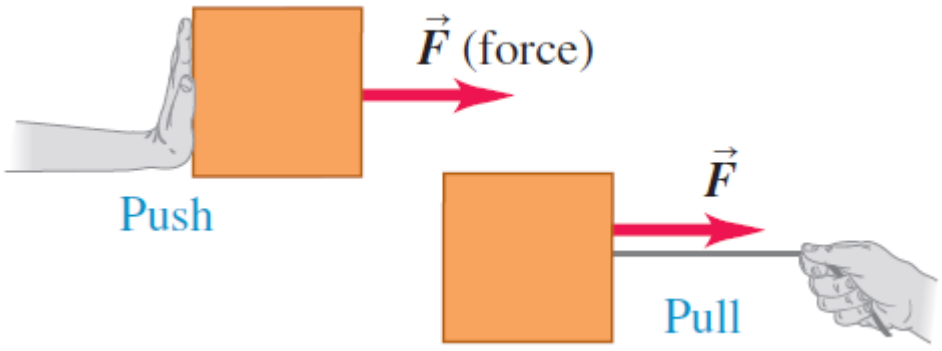
Law	Physical Quantity	Real-World Application
First Law	Inertia	Seat belts in cars
Second Law	Force, Mass, Acceleration	Rocket propulsion
Third Law	Action-Reaction Pairs	Walking, swimming

Newton’s Laws of Motion (Limitation)

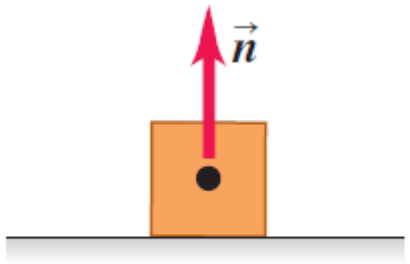
Limitation	Domain of Failure	Correct Theory
High Speeds (≈ speed of light)	Relativistic Domain	**Special Relativity**
Very Small Sizes (atoms, particles)	Quantum Domain	**Quantum Mechanics**
Gravitational Fields (strong fields, cosmology)	Cosmological Scale	**General Relativity**
Accelerating Reference Frames	Non-Inertial Frames	Requires **Fictitious Forces**
Origin of Forces	Fundamental Physics	**Standard Model**, **GR**

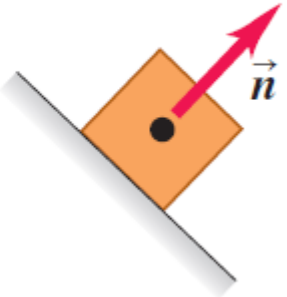
Force and Interactions

A force is a push or a pull. A force is an interaction between two objects or between an object and its environment.



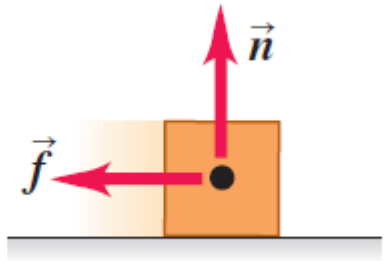
Normal force $\vec{n}$: When an object rests or pushes on a surface, the surface exerts a push on it that is directed perpendicular to the surface.



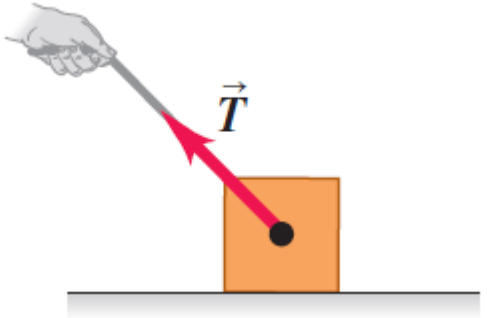


Force and Interactions

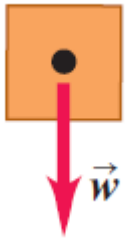
Friction force $\bar{f}$: In addition to the normal force, a surface may exert a friction force on an object, directed parallel to the surface.



Tension force $\bar{T}$: A pulling force exerted on an object by a rope, cord, etc

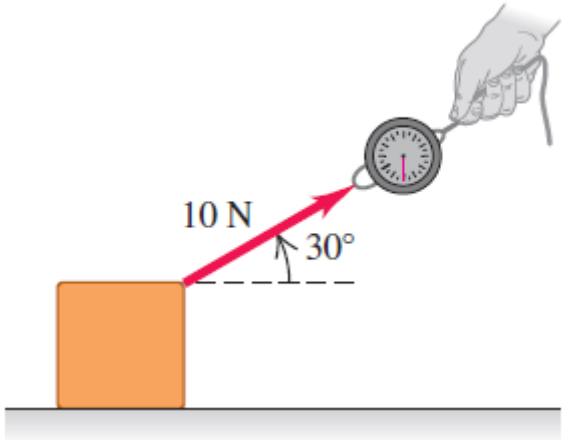


Weight $\bar{w}$: The pull of gravity on an object is a long-range force.

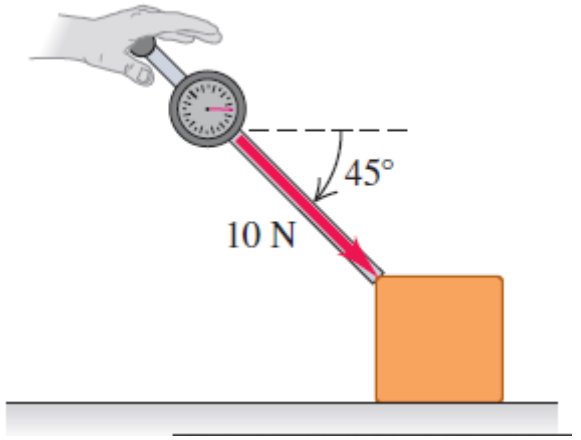


Force and Interactions

A 10 N pull directed 30 above the horizontal

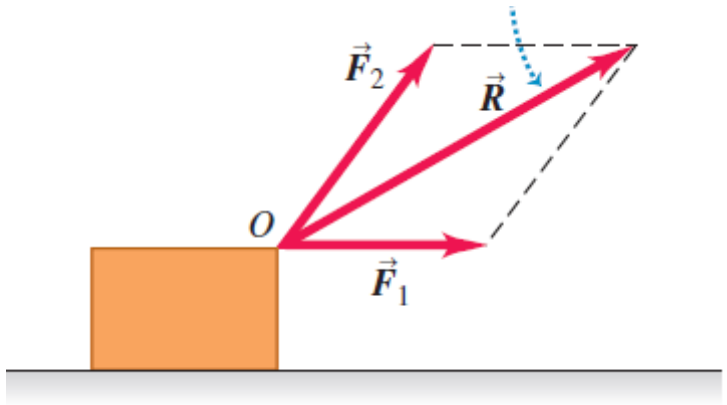


A 10 N push directed 45 below the horizontal

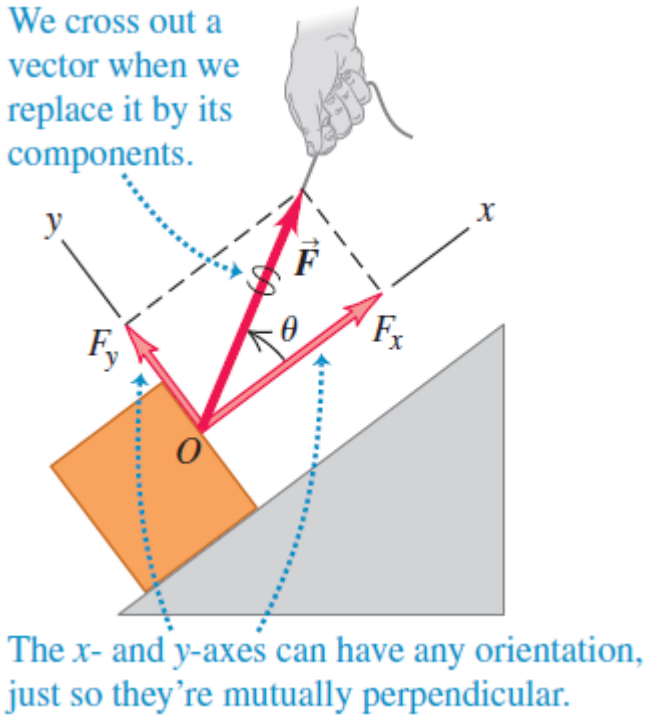


Superposition of Force

Two forces $\overline{\boldsymbol{F_1}}$ and $\overline{\boldsymbol{F_2}}$ acting on an object at point *O* have the same effect as a single force $\overline{\boldsymbol{R}}$ equal to their vector sum.



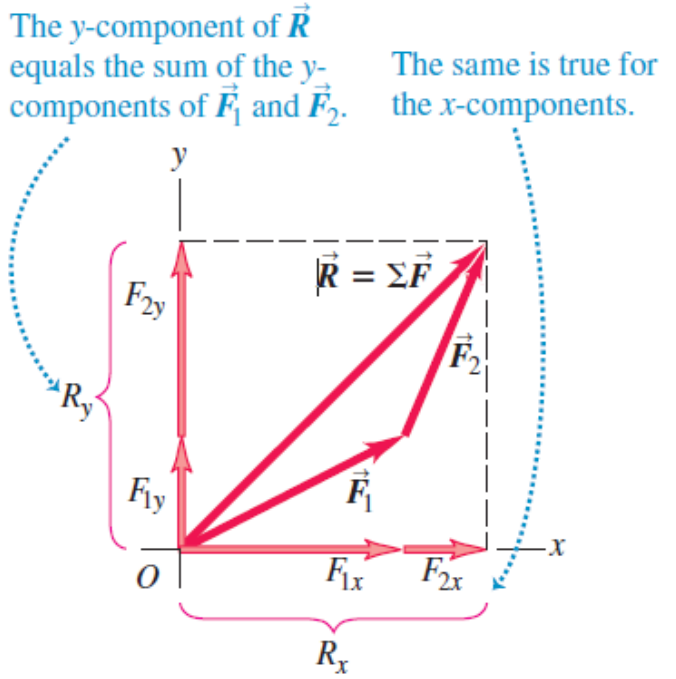
F_x and F_y are the components of $\bar{F}$ parallel and perpendicular to the sloping surface of the inclined plane.



Superposition of Force

Finding the components of the vector sum $\overline{\boldsymbol{R}}$ of two forces $\overline{\boldsymbol{F_1}}$ and $\overline{\boldsymbol{F_2}}$.

$$\overline{\boldsymbol{R}} = \overline{\boldsymbol{F_1}} + \overline{\boldsymbol{F_2}} + \overline{\boldsymbol{F_3}} + \cdots$$



$$R_x = \sum F_x$$

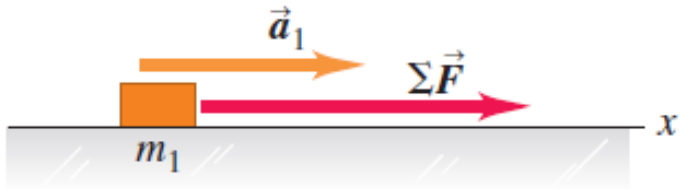
$$R_y = \sum F_y$$

$$\mathrm{R} = \sqrt{R_y + R_x}$$

$$\theta = tan^{-1}(R_y/R_x)$$

Newton's Laws

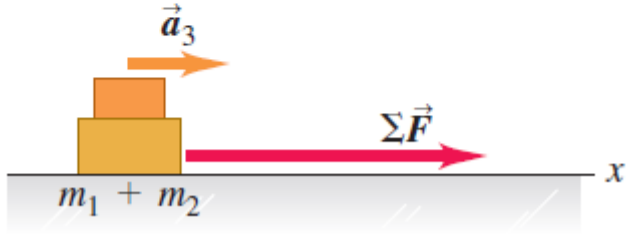
(a) A known net external force $\Sigma\vec{F}$ causes an object with mass m_1 to have an acceleration $\vec{a}_1$.



(b) Applying the same net external force $\Sigma\vec{F}$ to a second object and noting the acceleration allow us to measure the mass.



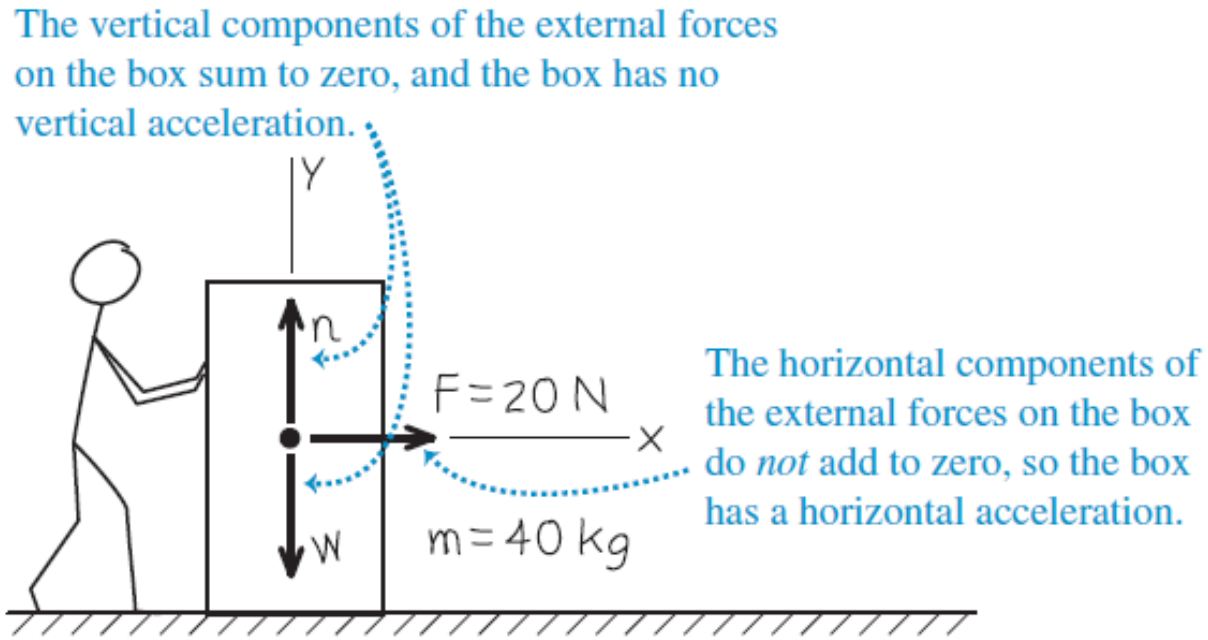
(c) When the two objects are fastened together, the same method shows that their composite mass is the sum of their individual masses.



Example

A worker applies a constant horizontal force with magnitude 20 N to a box with mass 40 kg resting on a level, freshly waxed floor with negligible friction. What is the acceleration of the box?

- **Vertical forces:** Weight mg downward and normal force N upward — they cancel each other (no vertical acceleration).
- **Horizontal force:** The applied force $F = 20$ Nis the only horizontal force.



Example

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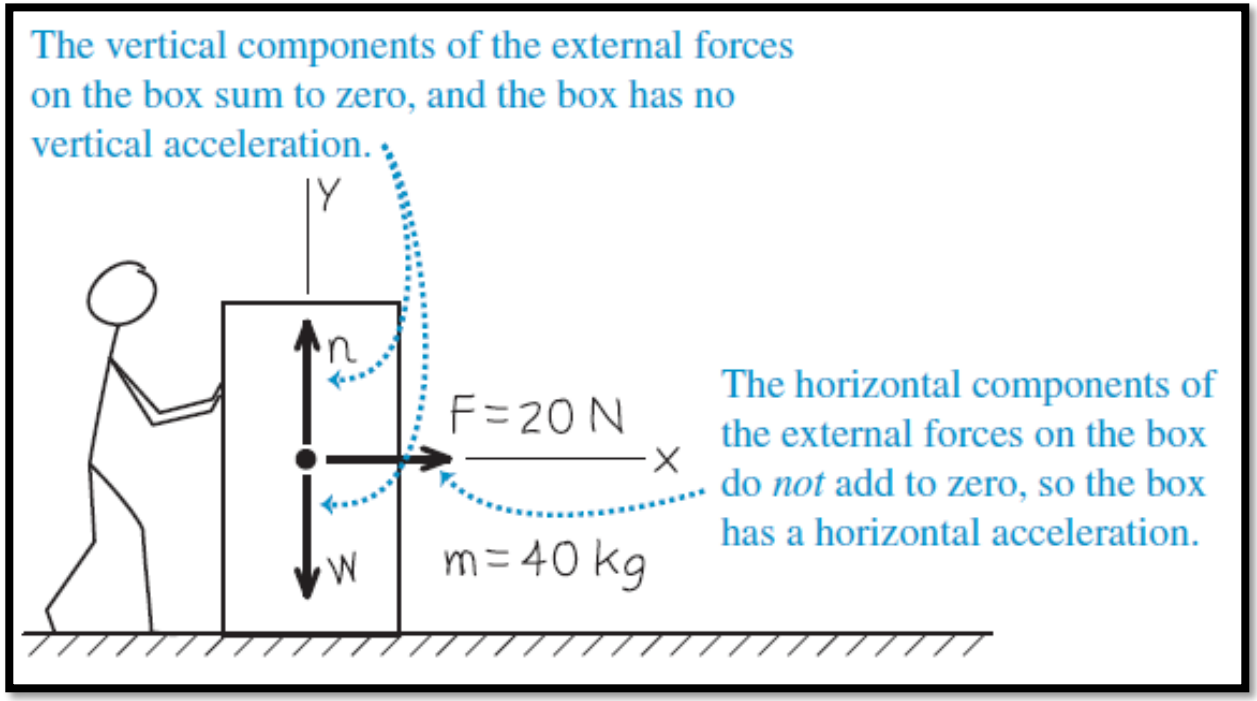
$$\sum \overline{F_x} = m\bar{a}_x = 20N$$

$$\sum \overline{F_y} = m\bar{a}_y = 0\ N$$

$$a = [\sum F]/m = \frac{20}{40} = 0.5\ m/s^2$$

Acceleration is in the same direction as the applied force (horizontal).

$a = 0.5$ m/s^2 in the direction of the applied force



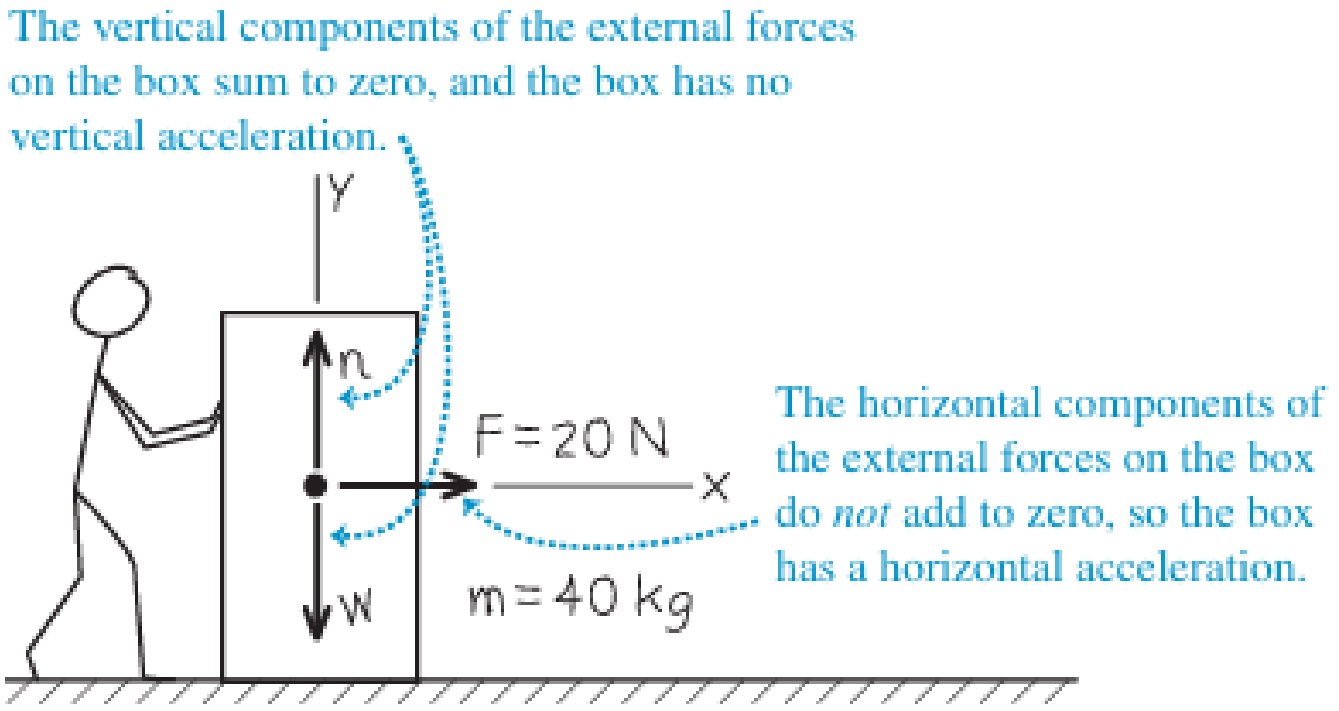
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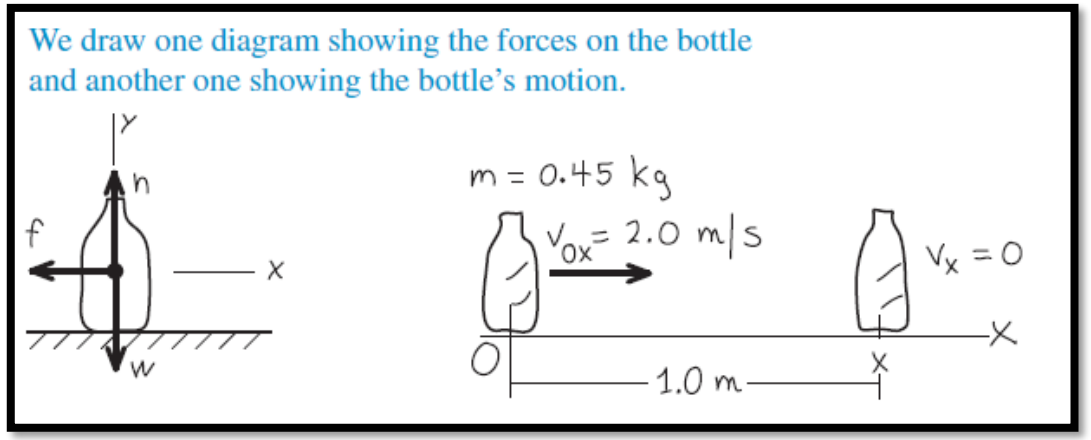


Example

A waiter shoves a ketchup bottle with mass $0.45\ kg$ to his right along a smooth, level lunch counter. The bottle leaves his hand moving at $2\ m/s$, then slows down as it slides because of a constant horizontal friction force exerted on it by the countertop. It slides for $1\ m$ before coming to rest. What are the magnitude and direction of the friction force acting on the bottle?

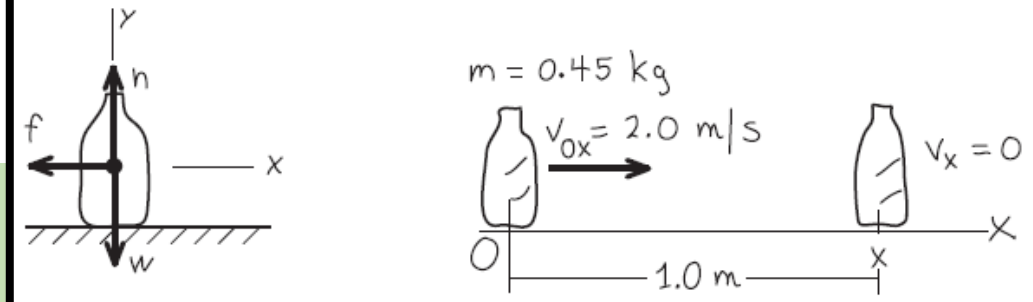
- Mass of the bottle: $m = 0.45$ kg
- Initial velocity: $v_0 = 2$ m/s(to the right)
- Final velocity: $v = 0$ m/s(comes to rest)
- Distance traveled: $d = 1$ m
- The surface is level (no vertical acceleration)
- Friction is the only horizontal force after the bottle leaves the hand

hand



Example

We draw one diagram showing the forces on the bottle and another one showing the bottle's motion.



Choose a Coordinate System

Let the positive x-direction be to the right (direction of initial motion). The bottle slows down, so the acceleration must be to the left (opposite to the velocity).

Find the Acceleration Using Kinematics

$$v_0 = 2 \text{ m/s}, v = 0 \text{ m/s}, d = 1 \text{ m}$$

Use the kinematic equation:

$$v^2 = v_0^2 + 2ad$$

Solve for a:

$$0 = (2)^2 + 2a(1)$$
$$0 = 4 + 2a$$
$$2a = -4$$
$$a = -2 \text{ m/s}^2$$

The negative sign means the acceleration is directed to the left (opposite to the initial motion).

Apply Newton's Second Law

The only horizontal force acting on the bottle is the friction force f. So:

$$F_{\text{net}} = ma$$
$$-f = ma$$
$$f = -ma$$

$$f = -(0.45)(-2)$$
$$f = 0.9 \text{ N}$$

The positive value indicates the magnitude. The direction is opposite to the motion — i.e., to the left.

- **Car A** is 1 km **East** of the intersection. Its position is (1, 0) km.
- **Car B** is 1 km **North** of the intersection. Its position is (0, 1) km.

- **Car A**: Moving East at 60 km/h → $\vec{v}_A = (60, 0)$ km/h
- **Car B**: Moving North at 80 km/h → $\vec{v}_B = (0, 80)$ km/h

Car A moves east at 60 km/h, and Car B moves north at 80 km/h. At a given moment, Car A is 1 km east of an intersection, and Car B is 1 km north of the intersection. Find

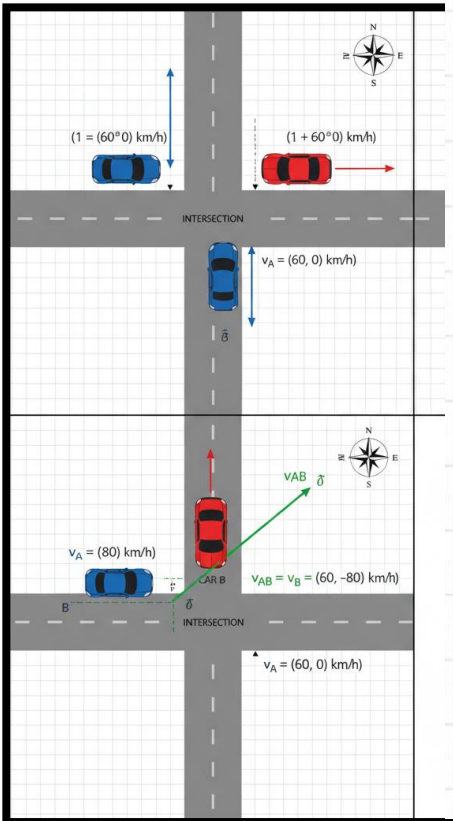
- Velocity of Car A relative to Car B

- Car A's velocity: $\vec{v}_A = (60, 0)\,\mathrm{km/h}$
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- Relative velocity of A with respect to B:
  - $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = (60, -80)\,\mathrm{km/h}$
- Magnitude: $|\vec{v}_{AB}| = \sqrt{60^2 + (-80)^2} = 100\,\mathrm{km/h}$
- Direction: $\theta = \tan^{-1}\left(\frac{80}{60}\right) \approx 53.13^\circ$ south of east

Exam

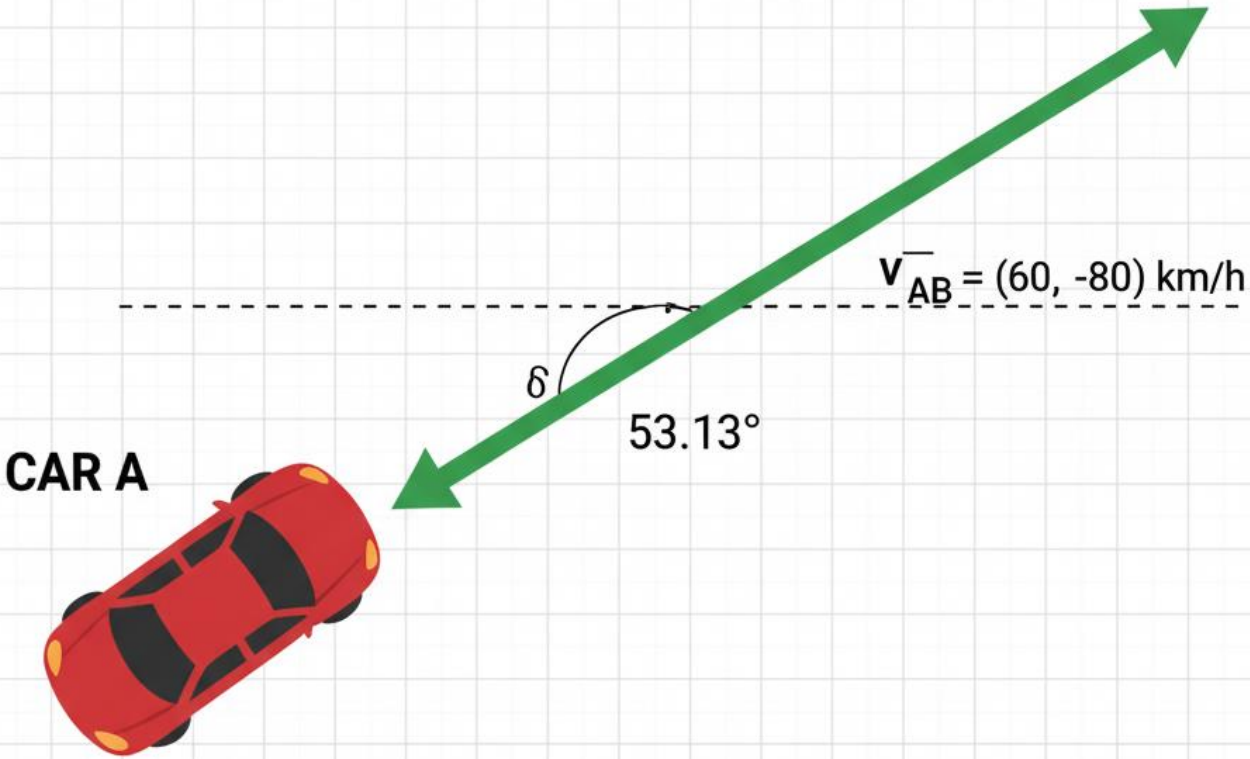
Car A moves ea ...given moment, Car A is 1 km east of an ...n. Find

- Velocity of C



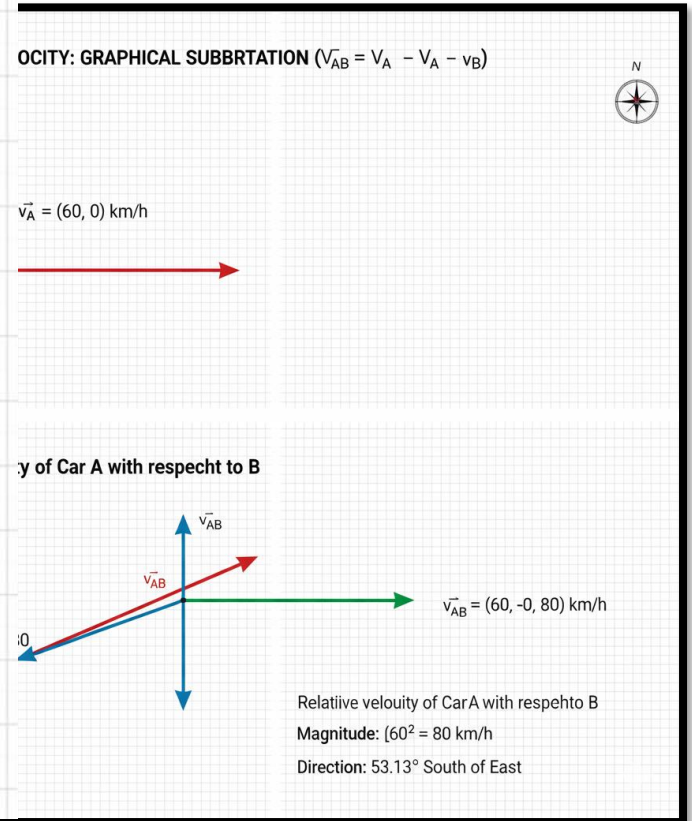
RELATIVE VELOCITY OF CAR A WITH RESPECT B





Magnitude: $[\bar{v}_{AB}] = \sqrt{(60^2 + -80)^2} = 100$ km/h

Direction: $\delta = \arctan(80/60) \approx 53.13°$ South of East



Example

A boat aims directly across a 100 m wide river. The boat's speed relative to water is 4 m/s, and the river flows at 3 m/s. Find

- Boat's velocity relative to the ground
- Time to cross the river
- Downstream drift

- Let the river flow be in the x-direction and crossing be in the y-direction.
- Boat's velocity relative to water: $\vec{v}_{bw} = (0, 4)\,\mathrm{m/s}$
- River's velocity relative to ground: $\vec{v}_{wg} = (3, 0)\,\mathrm{m/s}$
- Boat's velocity relative to ground: $\vec{v}_{bg} = \vec{v}_{bw} + \vec{v}_{wg} = (3, 4)\,\mathrm{m/s}$
- Speed relative to ground: $|\vec{v}_{bg}| = \sqrt{3^2 + 4^2} = 5\,\mathrm{m/s}$
- Time to cross: $t = \frac{\text{width}}{v_y} = \frac{100}{4} = 25\,\mathrm{s}$
- Downstream drift: $x = v_x \cdot t = 3 \times 25 = 75\,\mathrm{m}$

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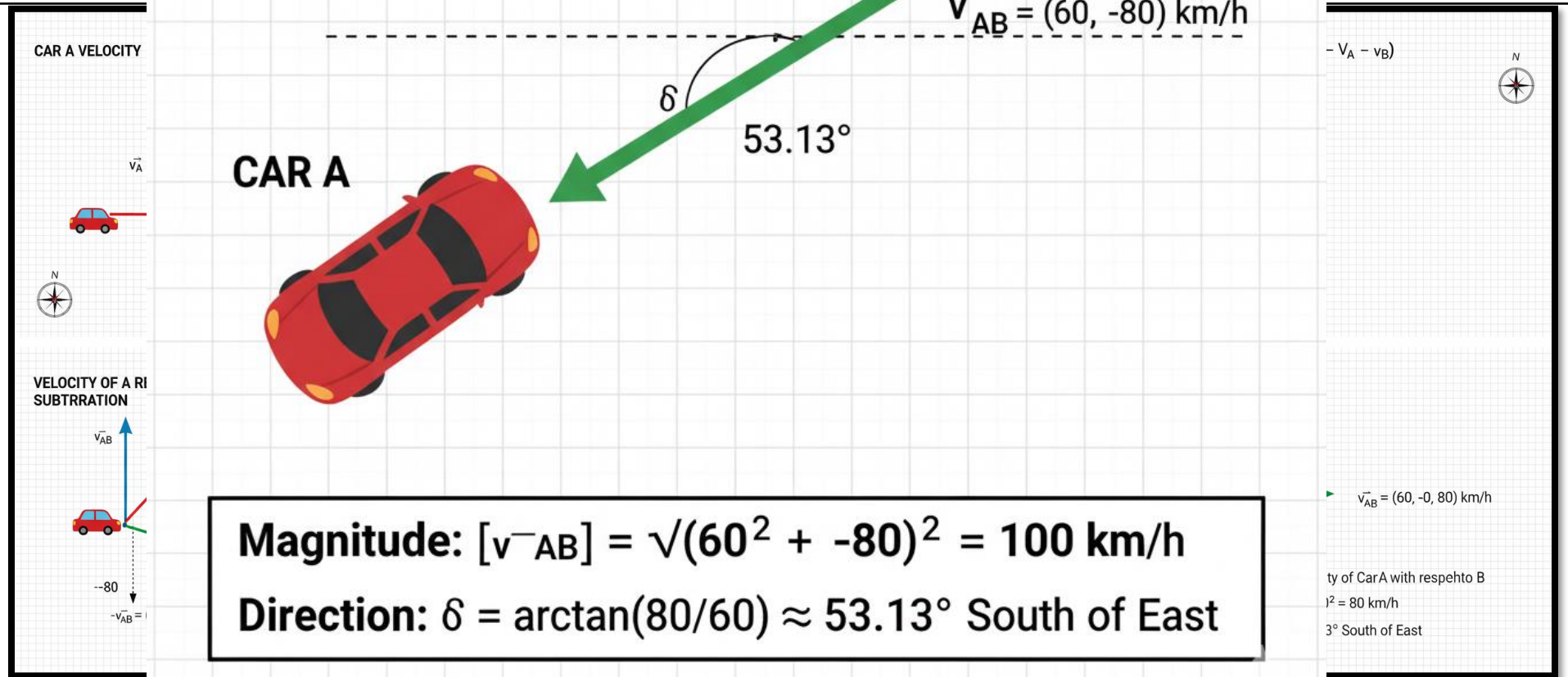
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Examp

Car A moves east
1 km east of an in
- Velocity of Car

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ind



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Net force determines how much an object accelerates.

$$\sum \bar{F} = m\bar{a}$$

Newton's second law: Each component of the net external force on an object ...

$$\sum F_x = ma_x \qquad \sum F_y = ma_y \qquad \sum F_z = ma_z$$

... equals the object's mass times the corresponding acceleration component.

ushing a empty shopping cart (small mass) vs. a full shopping cart (large mass) with the same force - the empty cart accelerates more.

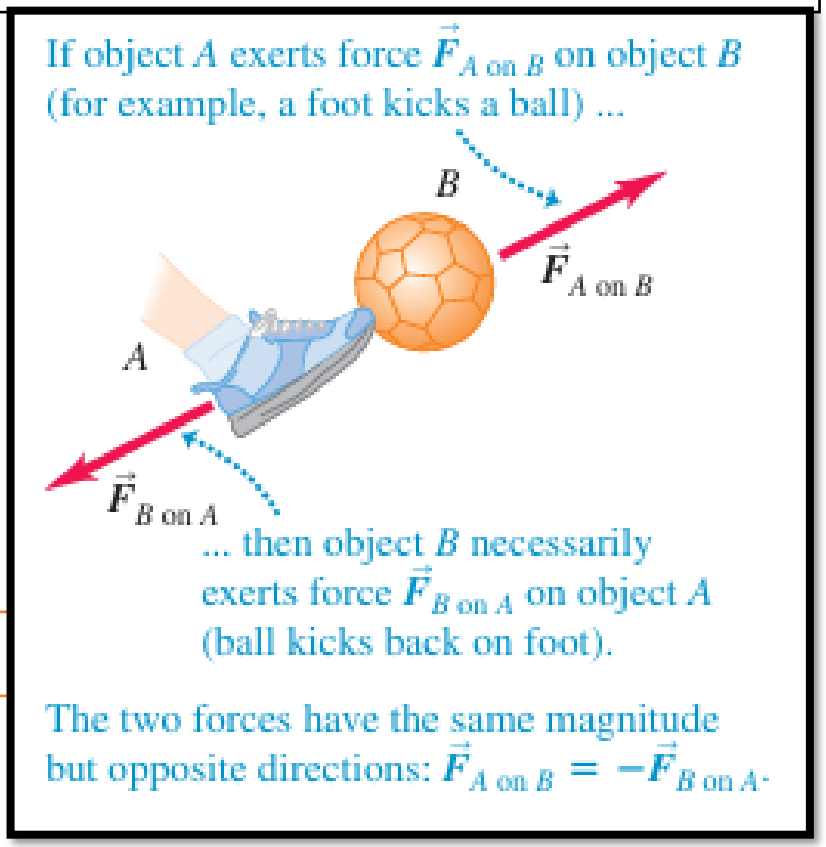
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Newton’s Laws of Motion

Law	Physical Quantity	Real-World Application
First Law	Inertia	Seat belts in cars
Second Law	Force, Mass, Acceleration	Rocket propulsion
Third Law	Action-Reaction Pairs	Walking, swimming

APPLICATION Blame Newton's Laws This car stopped because of Newton's second and third laws. During the impact, the car exerted a force on the tree; in accordance with the third law, the tree exerted an equally strong force back on the car. In accordance with the second law, the force of the tree on the car gave the car an acceleration that changed its velocity to zero.

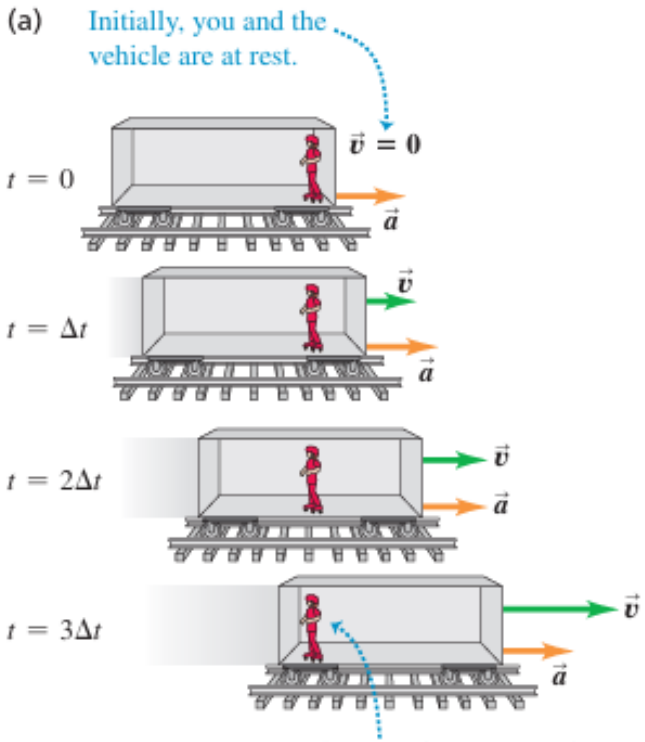


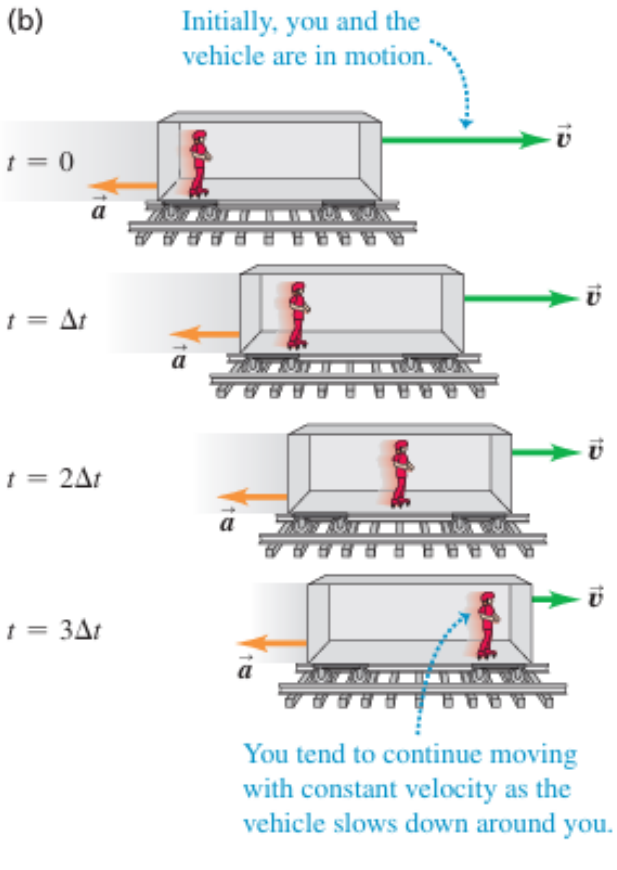
Newton's Laws of Motion (Limitation)

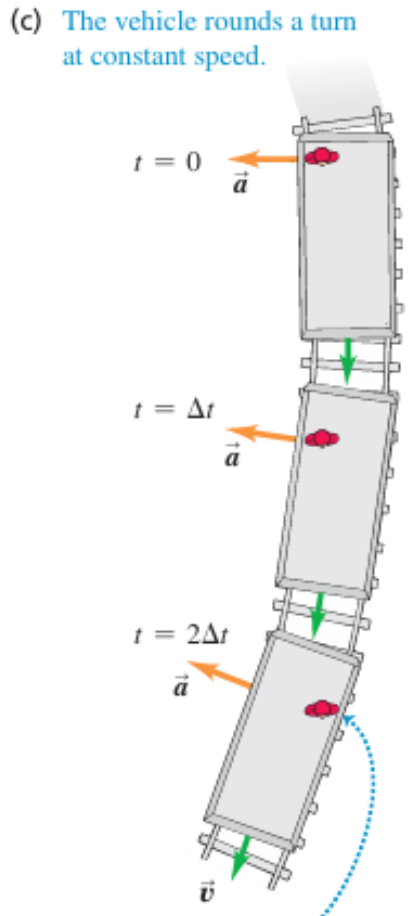
Limitation	**Domain of Failure**	**Correct Theory**
High Speeds (≈ speed of light)	Relativistic Domain	Special Relativity
Very Small Sizes (atoms, particles)	Quantum Domain	Quantum Mechanics
Gravitational Fields (strong fields, cosmology)	Cosmological Scale	General Relativity
Accelerating Reference Frames	Non-Inertial Frames	Requires Fictitious Forces
Origin of Forces	Fundamental Physics	Standard Model

Newton's Laws of Motion (Limitation)

Figure 4.10 Riding in an accelerating vehicle.



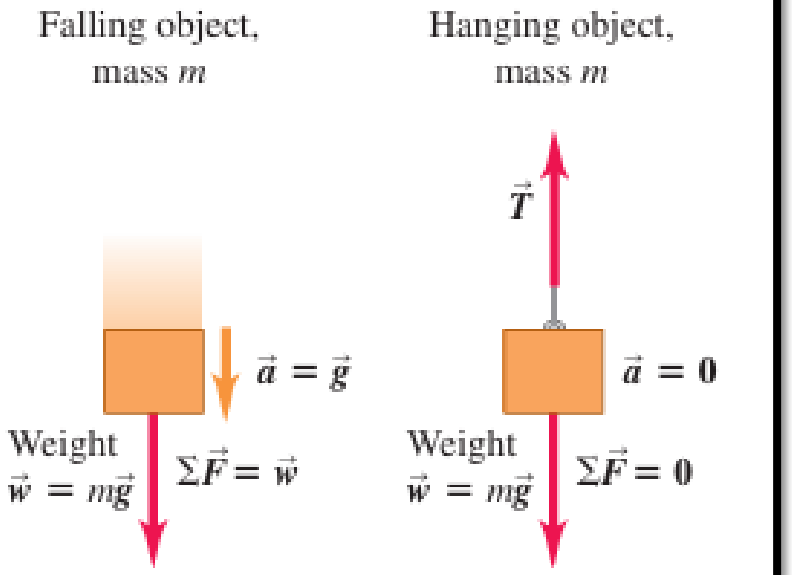




Weight of an object is the gravitational force that the earth exerts on the object.

Mass and Weight

Figure 4.20 Relating the mass and weight of an object.



- The relationship of mass to weight: $\vec{w} = m\vec{g}$.
- This relationship is the same whether an object is falling or stationary.

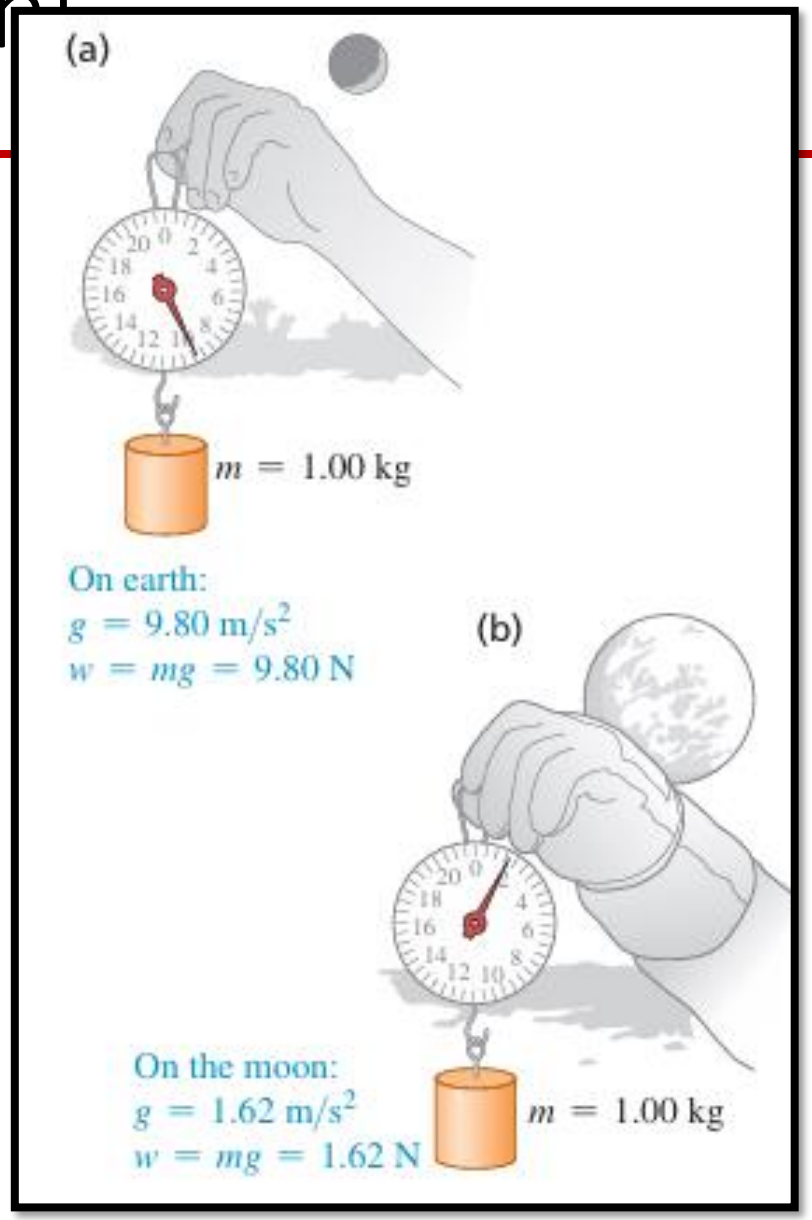
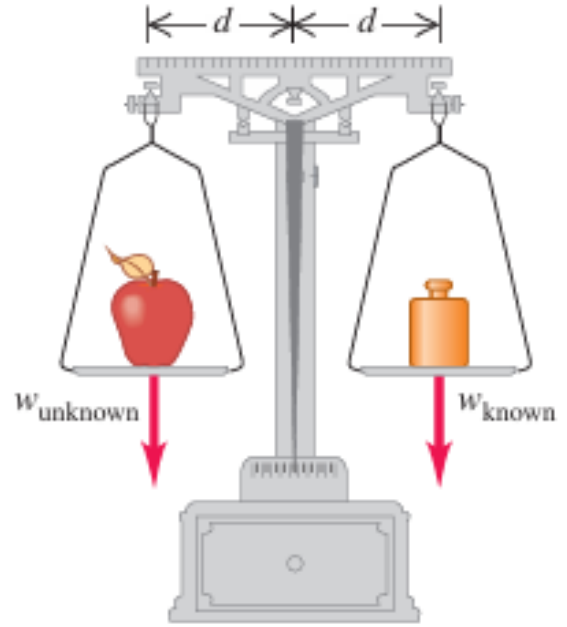
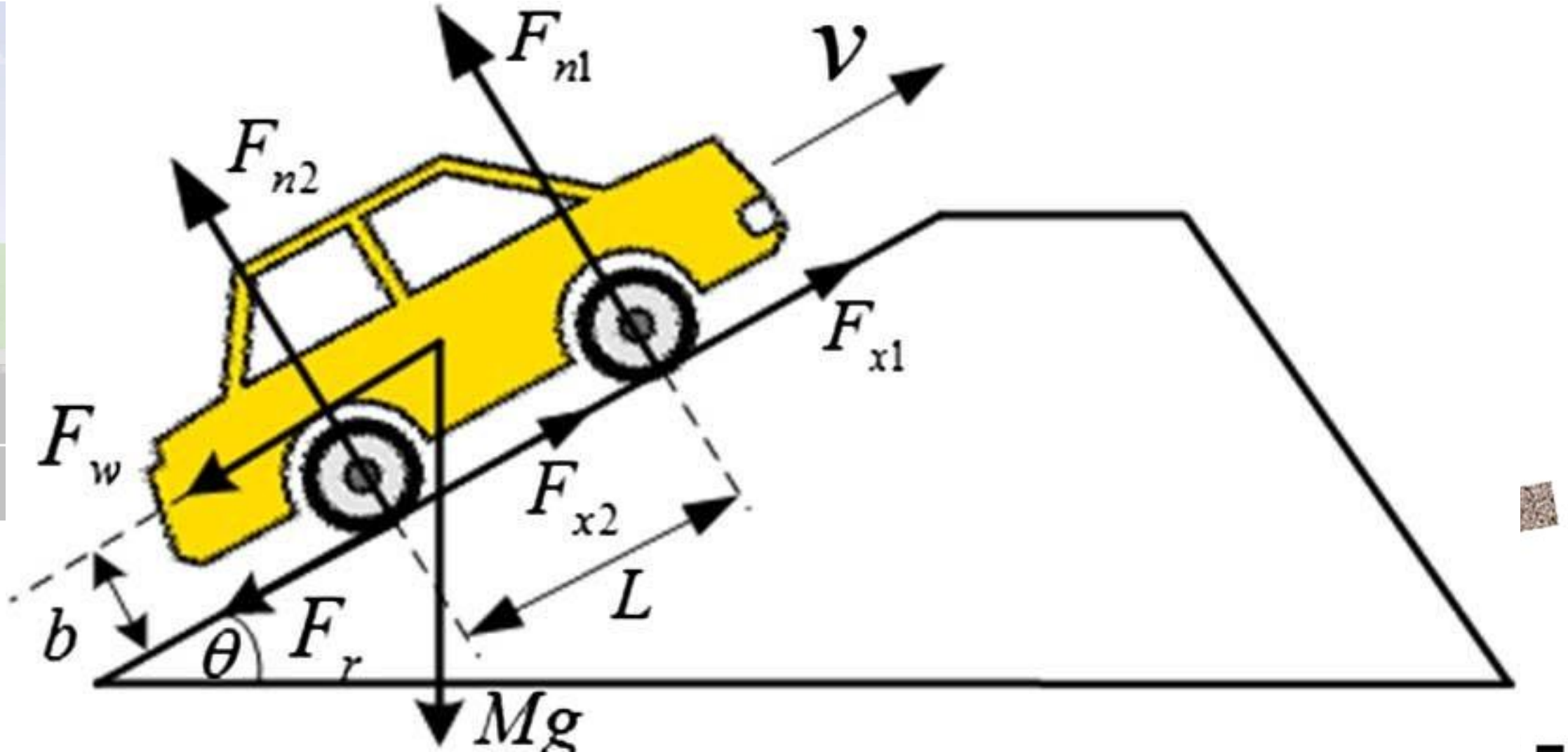


Figure 4.23 An equal-arm balance determines the mass of an object (such as an apple) by comparing its weight to a known weight.

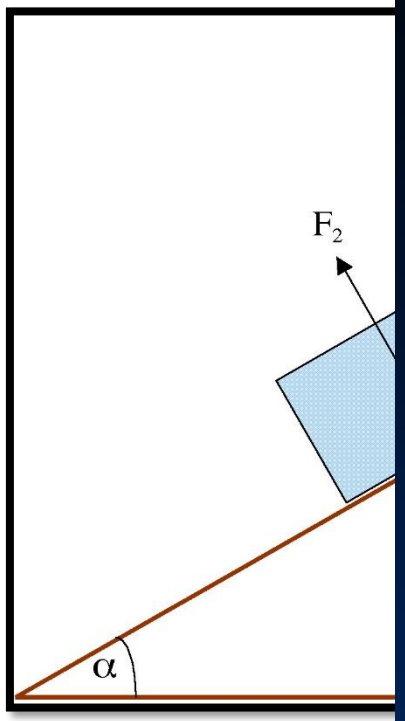


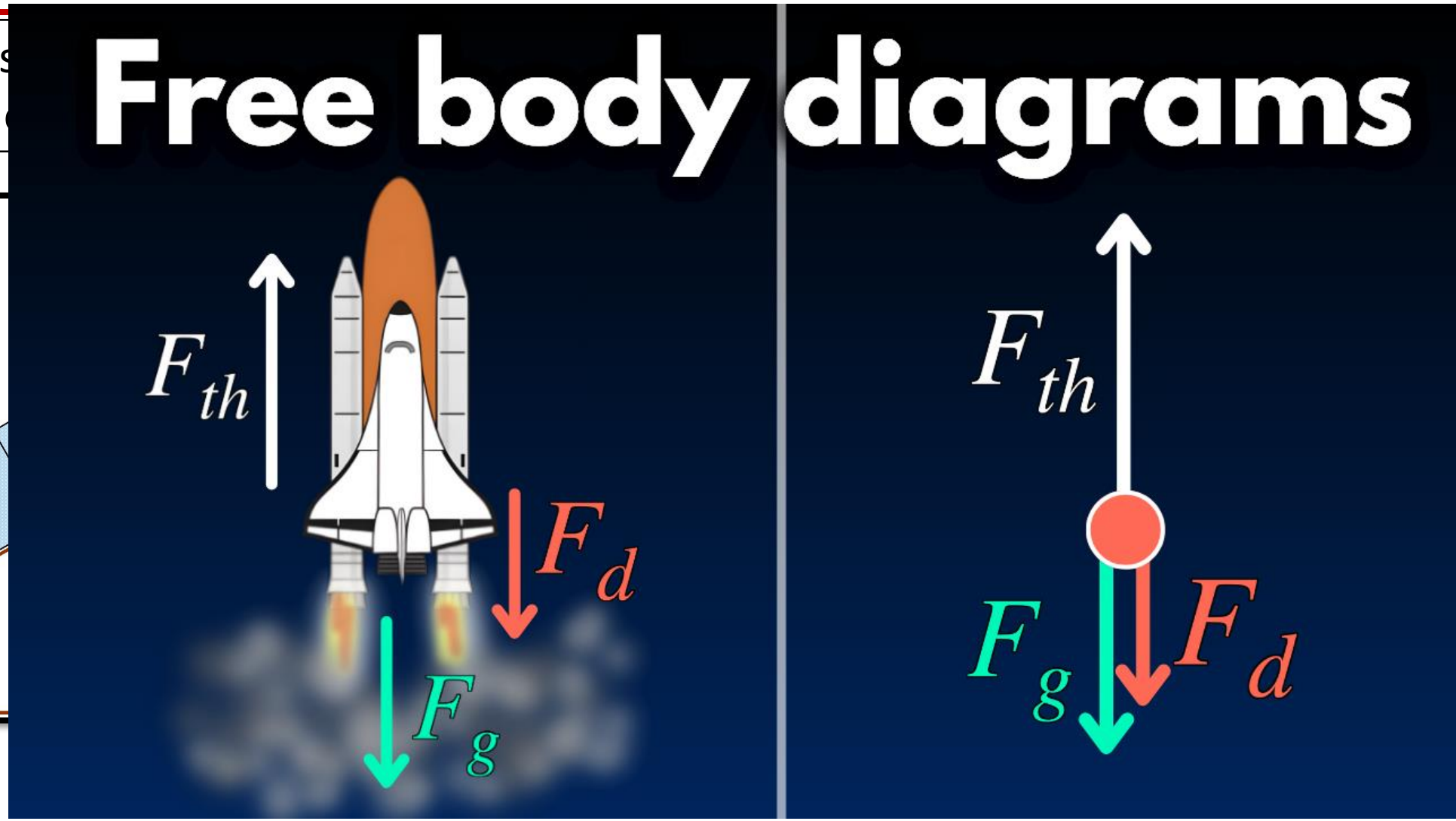
Free Body Diagram



Free Body Diagram

A free body diagram, is
and reactions exerted





Free Body Diagram

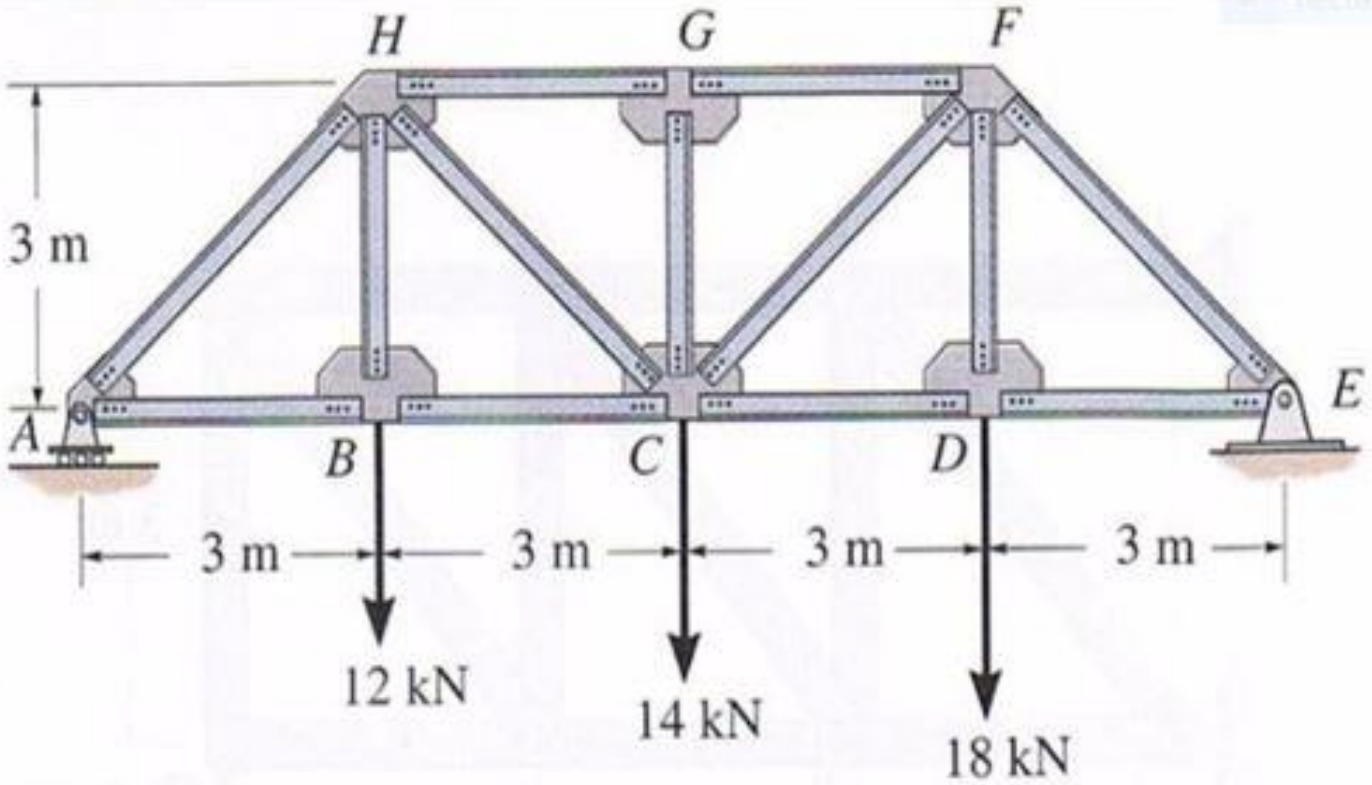
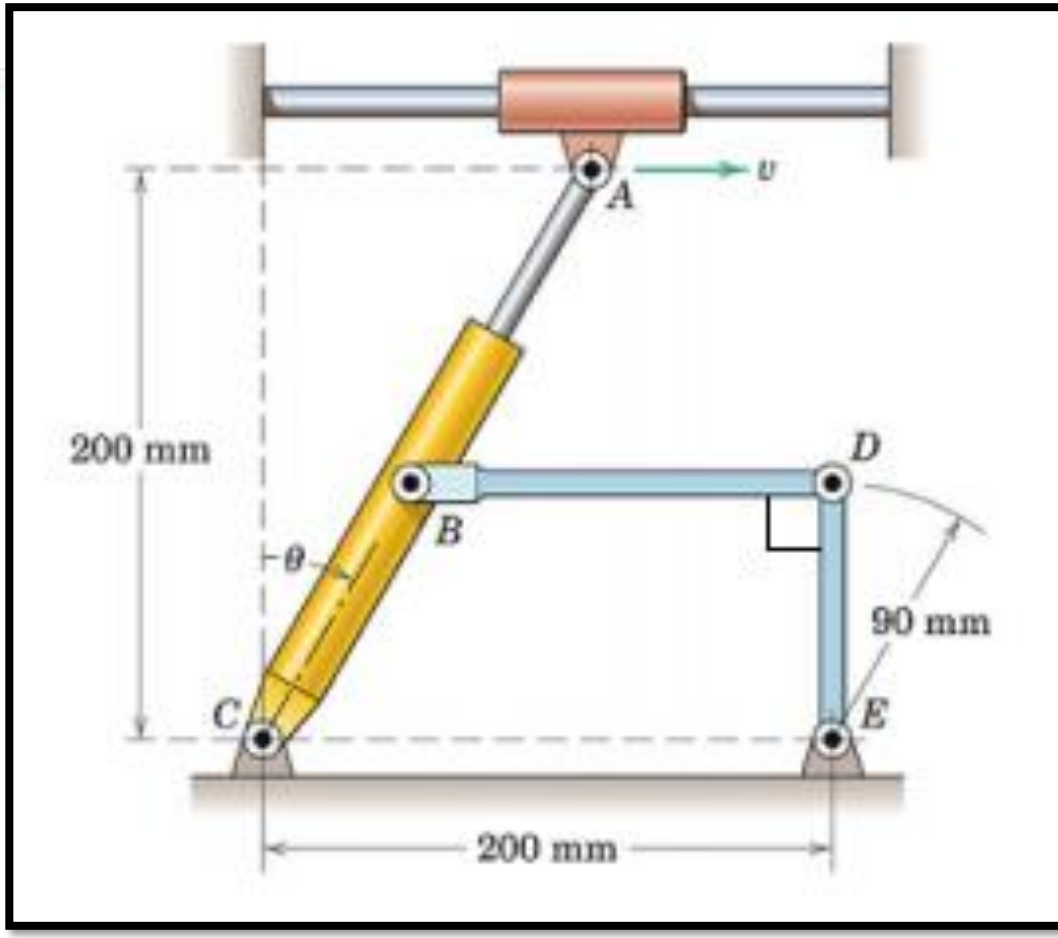
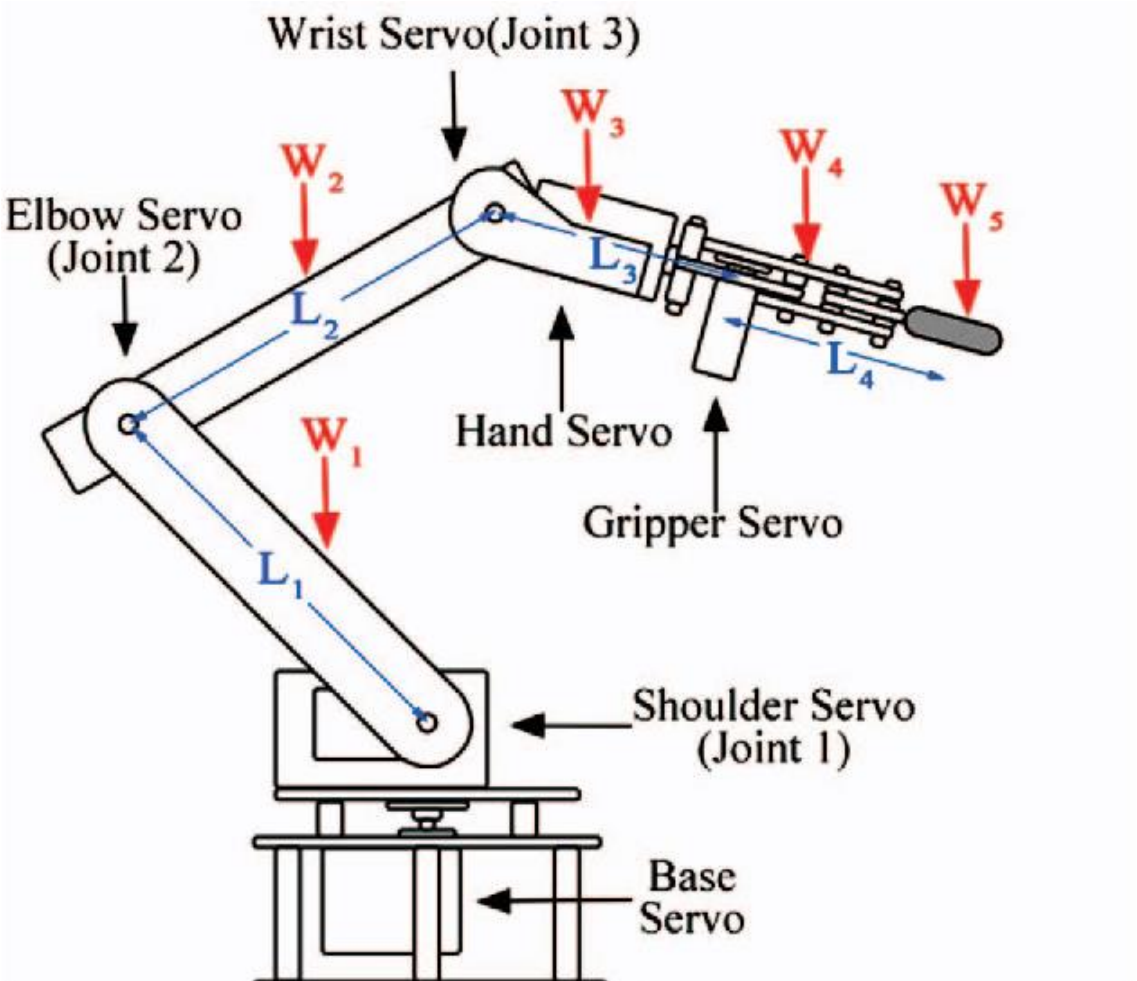
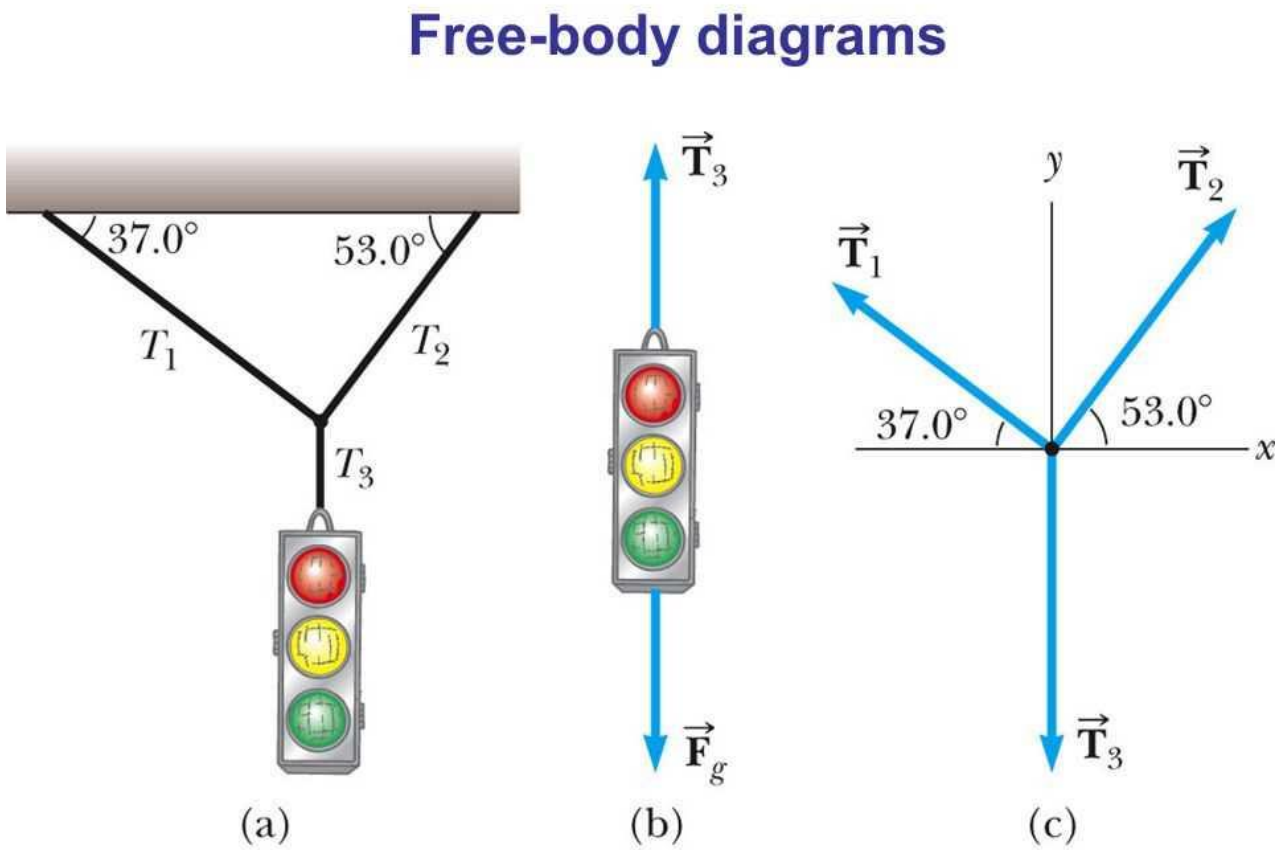


Figure QB1

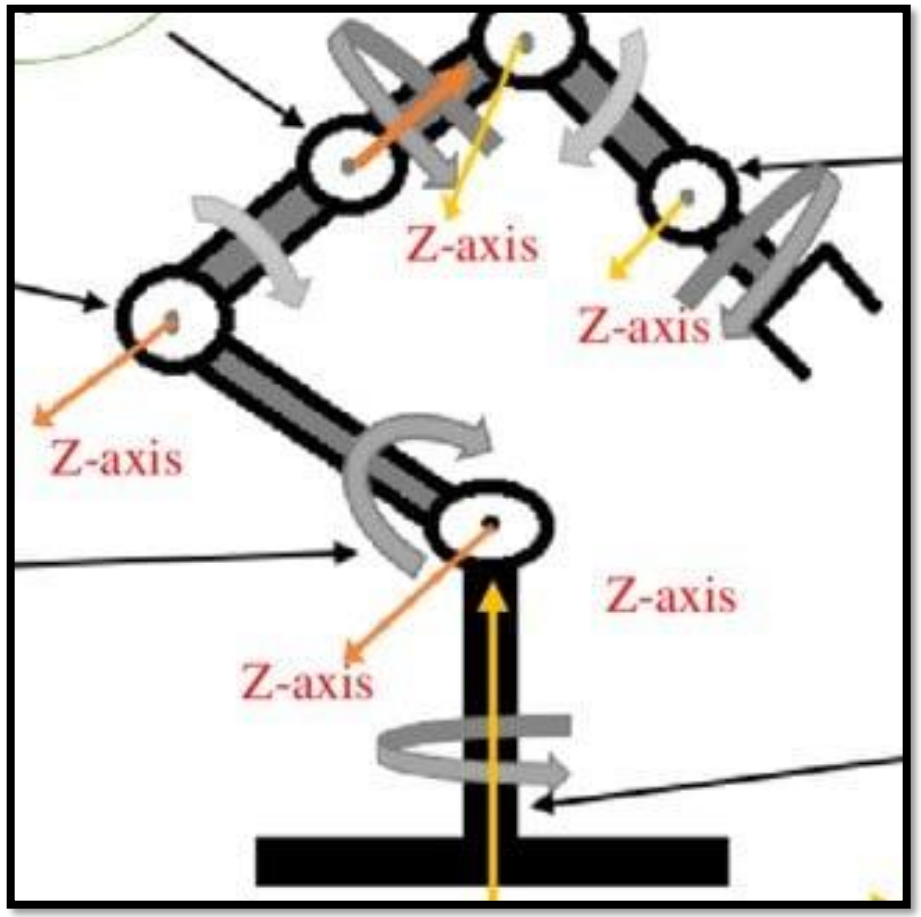


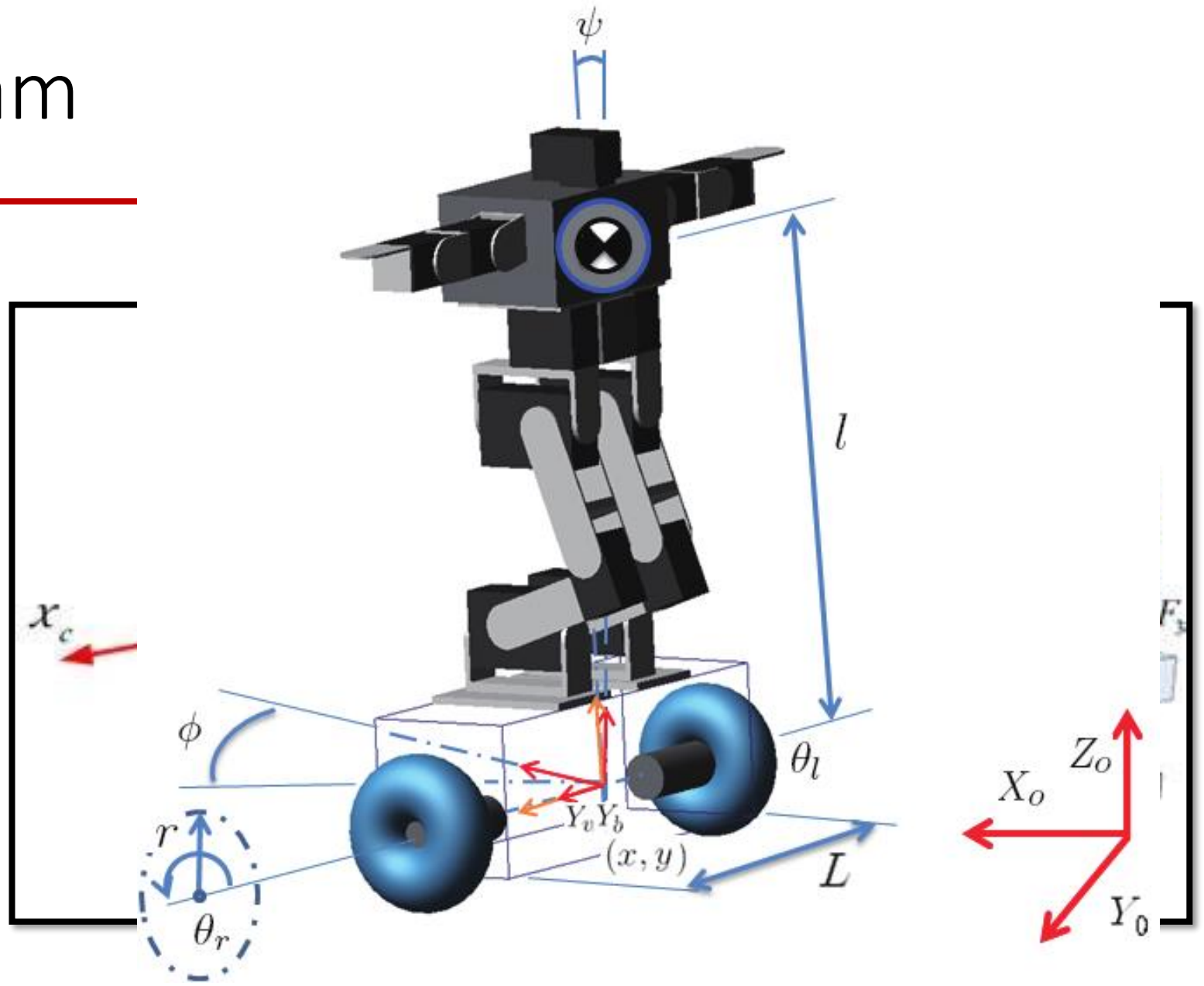
Free Body Diagram

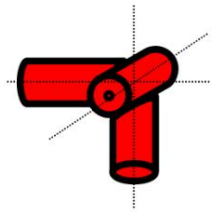




Free Body Diagram

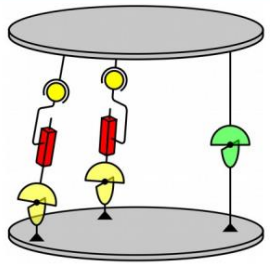






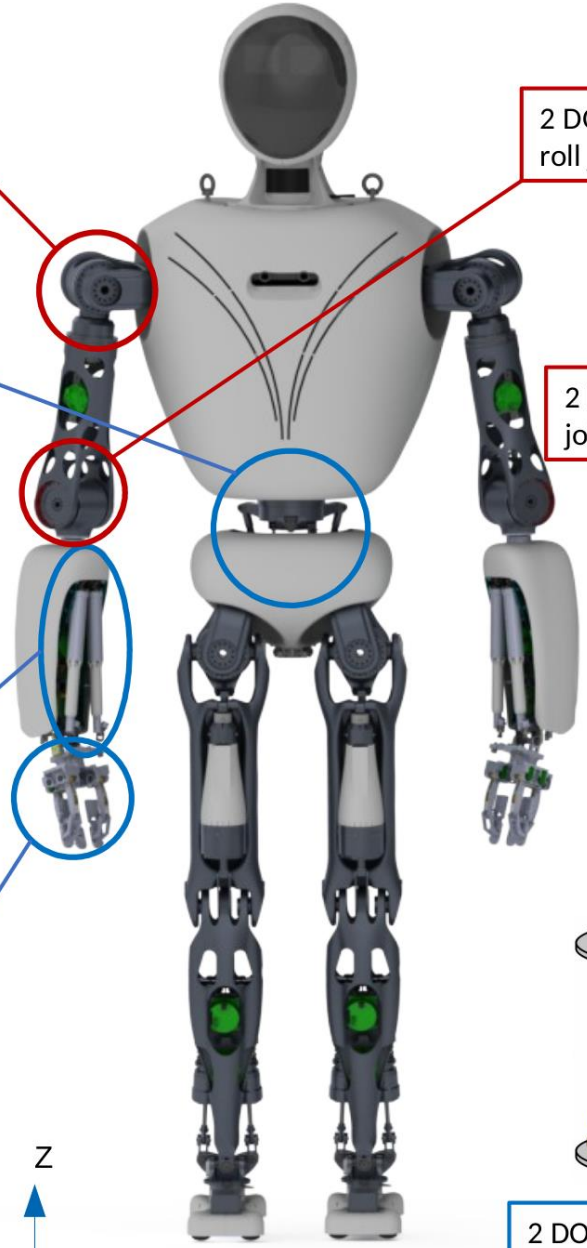
3 DOF shoulder joint, serially actuated

3 DOF torso joint, actuated by 2-SPU+1U mechanism (roll and pitch) and one rotary actuator in series (yaw)

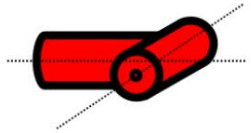


2 DOF wrist joint (pitch/yaw), actuated by 2-SPU+1U mechanism

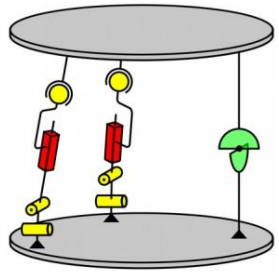
1 DOF underactuated, passive adaptive gripper



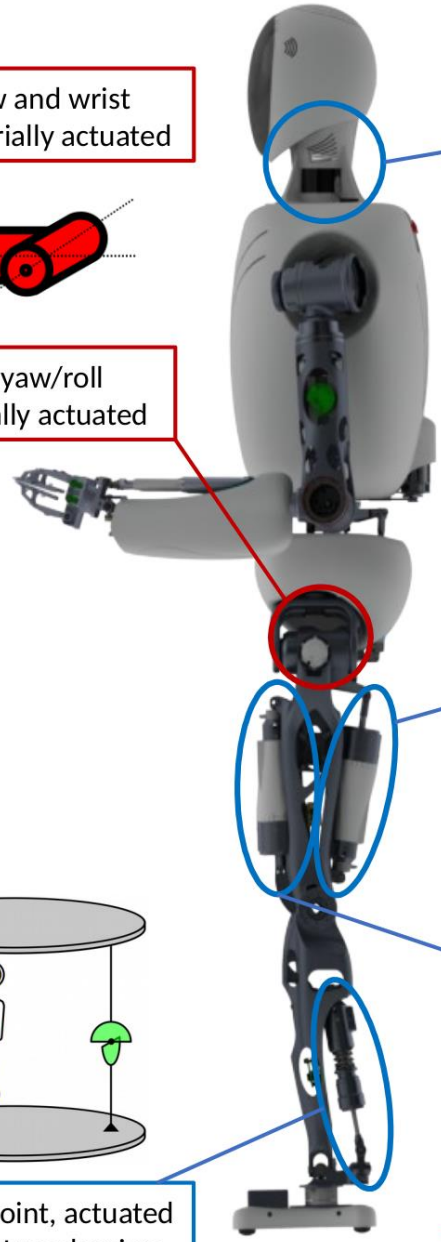
2 DOF elbow and wrist roll joint, serially actuated



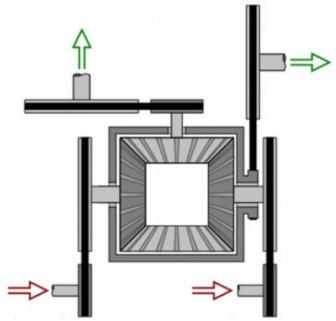
2 DOF hip yaw/roll joint, serially actuated



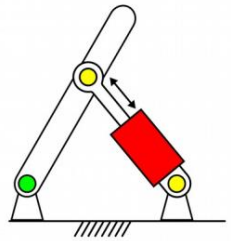
2 DOF ankle joint, actuated by 2-SPRR+1U mechanism



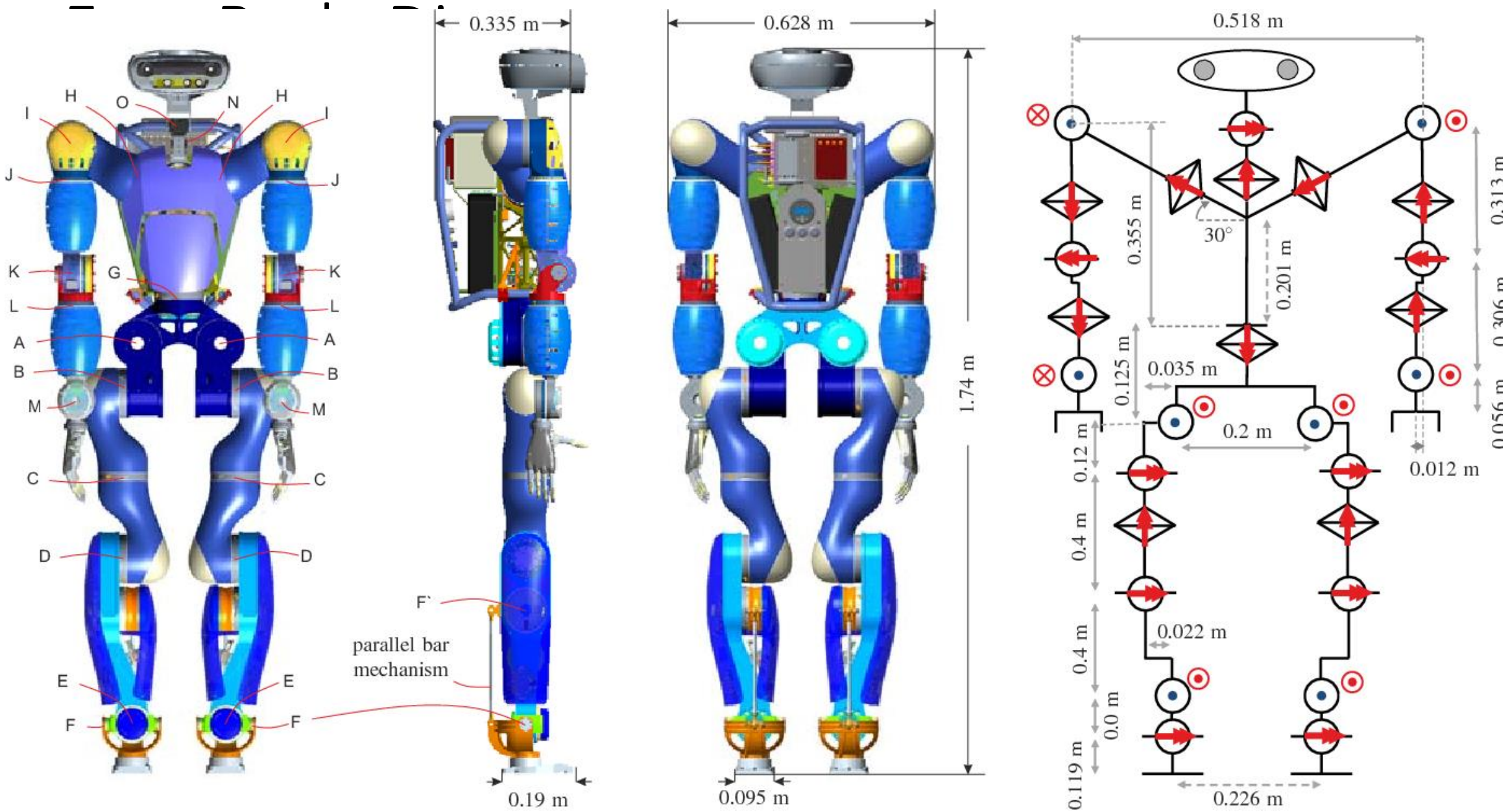
3 DOF head joint: 2 DOF differential mechanism + 1 DOF rotary actuator in series

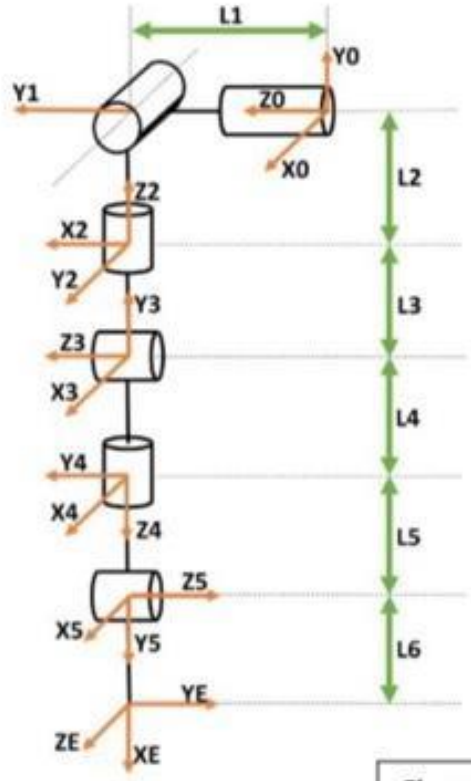


1 DOF hip pitch joint, actuated by 1-RRPR mechanism



1 DOF knee joint, actuated by 1-RRPR mechanism


H
O
N
H
I
I
J
J
K
G
K
L
L
A
A
B
B
M
M
C
C
D
D
E
E
F
F
0.335 m
F`
parallel bar
mechanism
0.19 m
0.628 m
1.74 m
0.095 m
0.518 m
0.355 m
30°
0.201 m
0.313 m
0.125 m
0.035 m
0.2 m
0.012 m
0.12 m
0.4 m
0.022 m
0.4 m
0.0 m
0.119 m
0.226 m



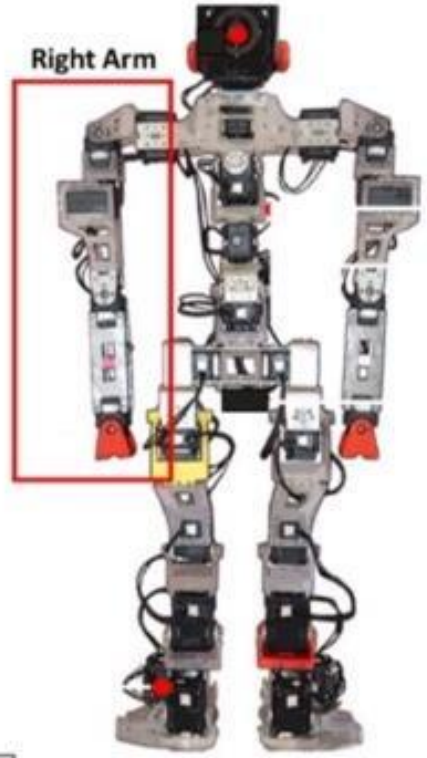


Figure 1

$$A_i = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i}c_{\alpha_i} & s_{\theta_i}s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i}c_{\alpha_i} & -c_{\theta_i}s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

DH matrix

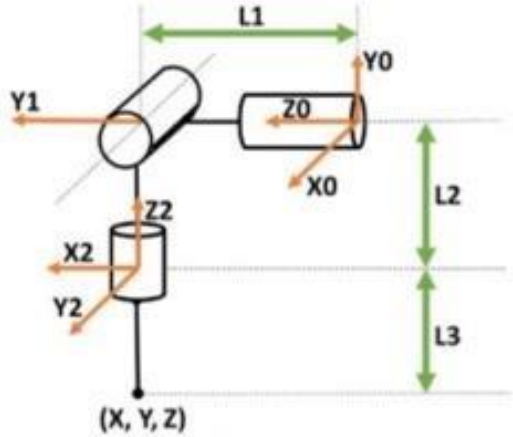
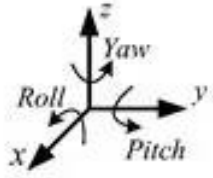
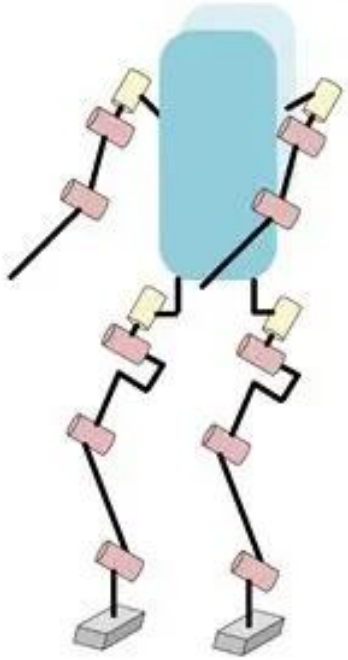


Figure 2









Free Body Diagram



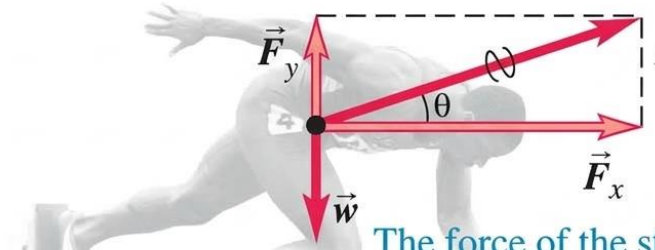


To jump up, this player is pushing down against the floor, increasing the upward reaction force $\vec{n}$ of the floor on him.

This player is a freely falling object.

(a)

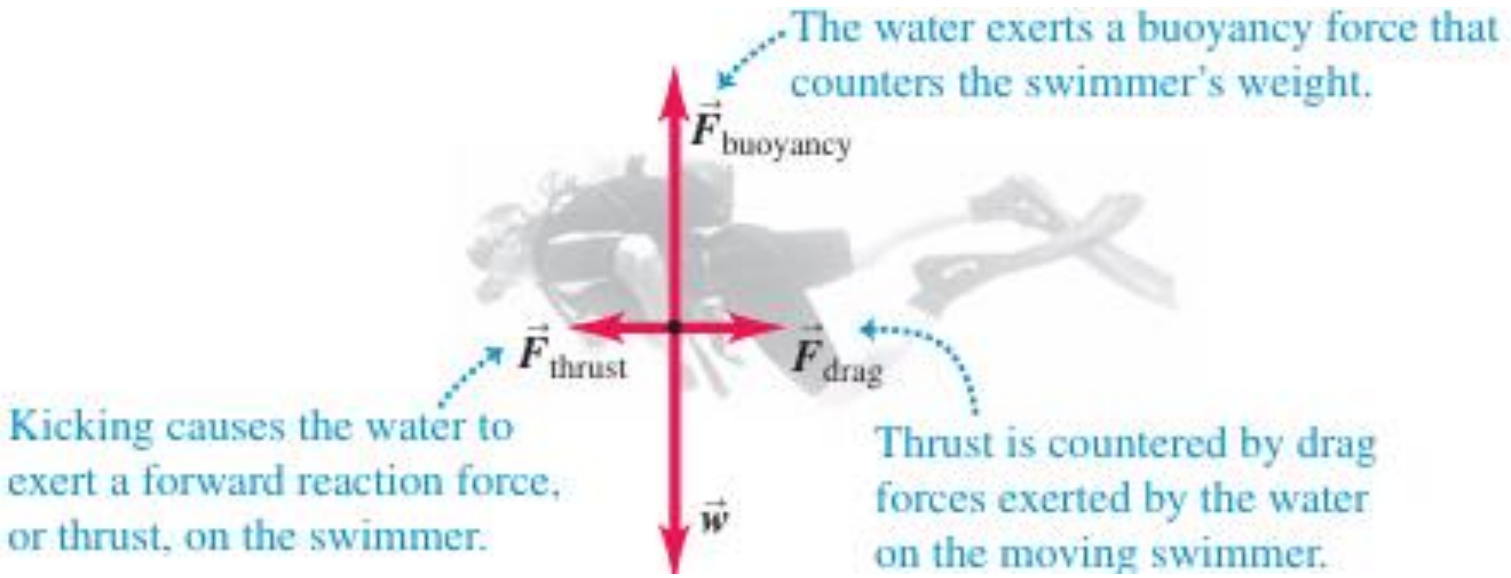




The force of the starting block on the runner has a vertical component that counteracts his weight and a large horizontal component that accelerates him.

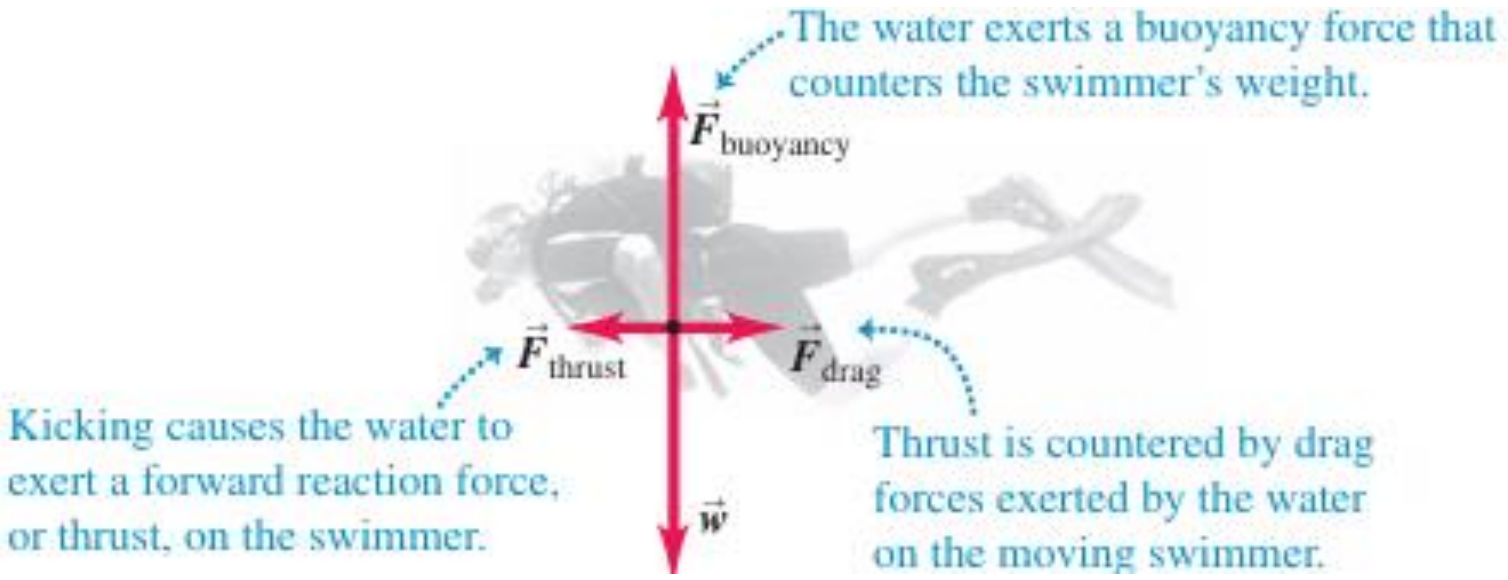
Free Body Diagram





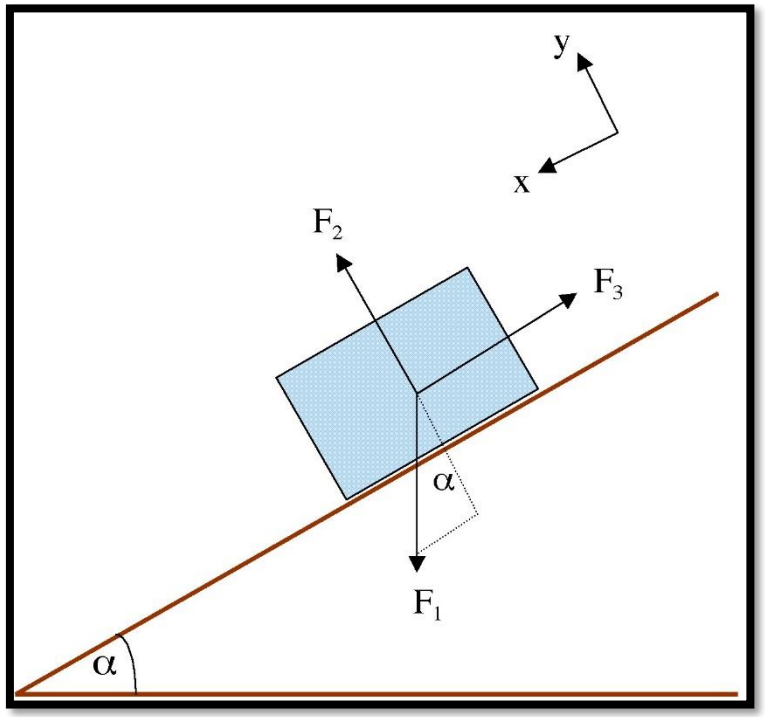
Free Body Diagram





Free Body Diagram

A free body diagram, is a sketch of an isolated body, which shows external forces on the body and reactions exerted on it by removing supports.

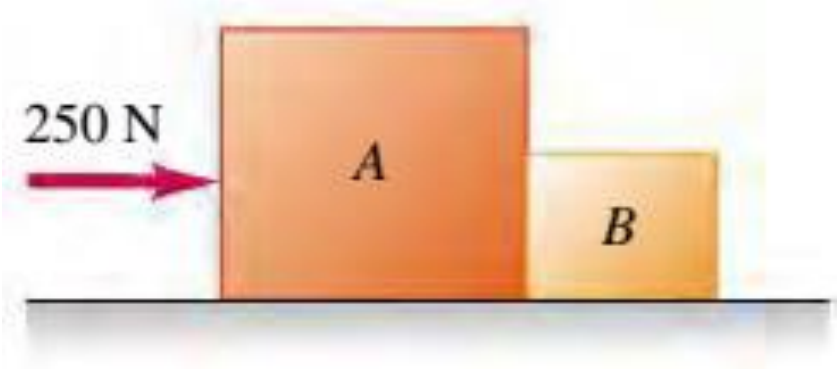


Example

Boxes *A* and *B* are in contact on a horizontal, frictionless Surface. Box *A* has **mass 20.0 kg** and box *B* has **mass 5.0 kg**. A horizontal **force of 250 N** is exerted on box *A*. **What is the magnitude of the force that box *A* exerts on box *B*?**

$$a_X = \frac{[\sum F]}{m_A + m_B} = \frac{250}{m_A + mB}$$

$$a_X = \frac{250}{25} = 10\ m/s^2$$

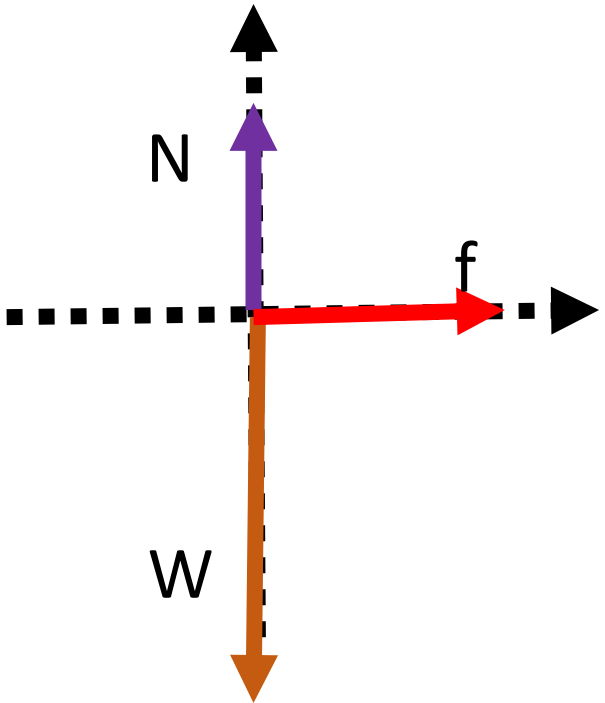


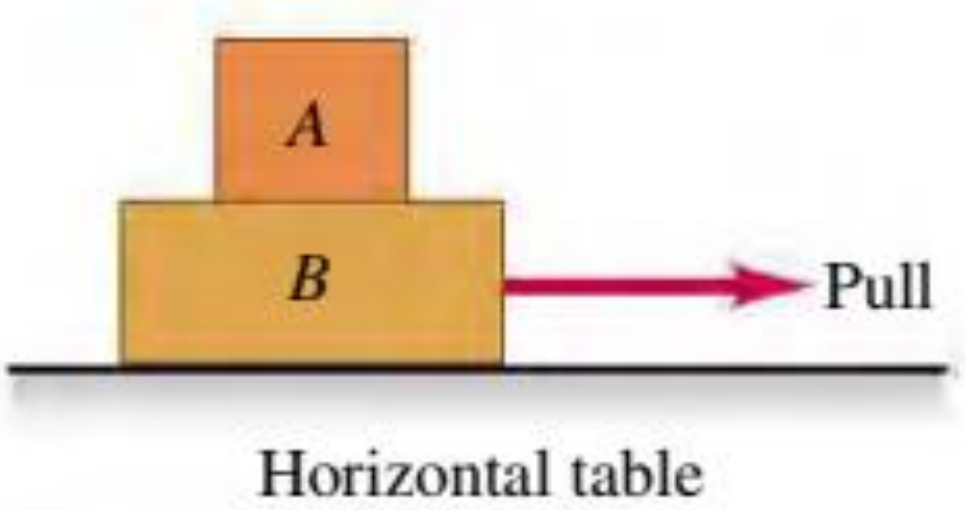
$$F_B = m_B a_X$$

$$F_B = (5)(10) = 50\ \text{N}$$

Example

You pull horizontally on block *B* in, causing both blocks to move together as a unit. For this moving system, **make a carefully labeled free-body diagram of block *A*** if (a) the table is frictionless and **(b)** there is friction between block *B* and the table and **the pull is equal in magnitude to the friction force** on block *B* due to the table.





Example

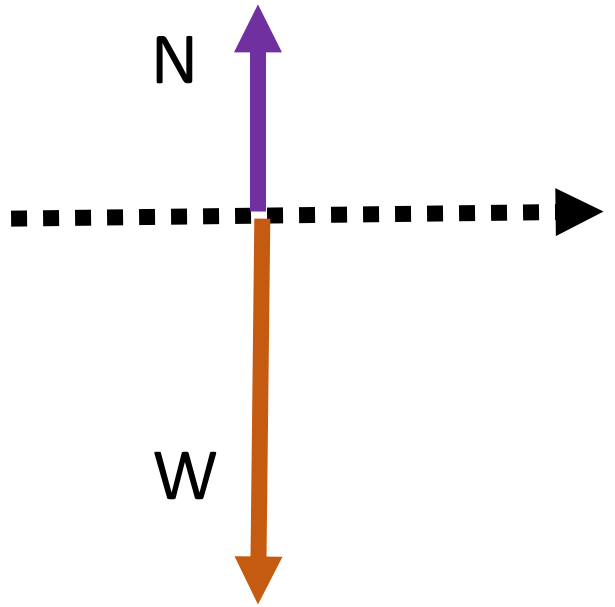
You walk into an elevator, step onto a scale, and push the "up" button. You recall that your normal **weight is 625 N**. **(a)** When the elevator has an upward **acceleration of magnitude 2.50** $\boldsymbol{m/s^2}$, what does the scale read? . **(b)** If you hold a **3.85-kg package** by a light vertical string, **what will be the tension** in this string when the elevator accelerates as in part (a)?

$$N - W = ma$$

$$N = W + ma$$

$$N = 625 + \left(\frac{W}{g}\right)2.5$$

$$N = 625 + \left(\frac{625}{9..8}\right)2.5 = 784\ N$$



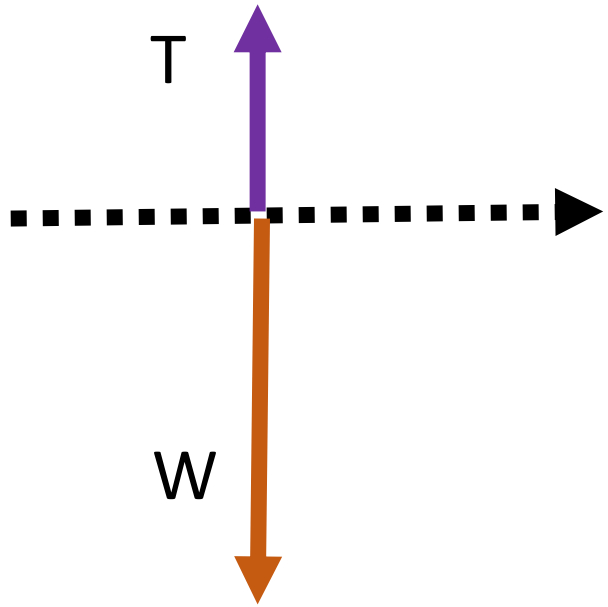
Example

You walk into an elevator, step onto a scale, and push the "up" button. You recall that your normal **weight is 625 N**. **(a)** When the elevator has an upward **acceleration of magnitude 2.50** $\boldsymbol{m/s^2}$, what does the scale read?

$$T - W = ma$$

$$T = W + ma$$

$$T = mg + ma$$

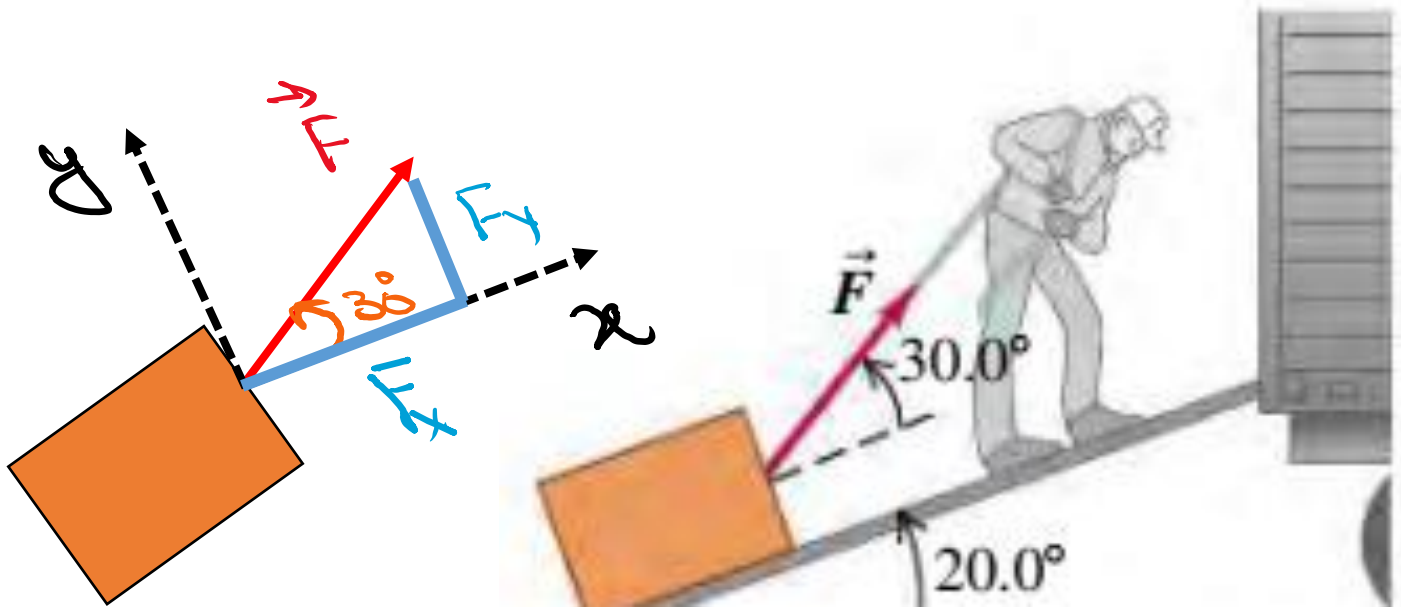


Example

A man is dragging a trunk up the loading ramp of a mover's truck. The ramp has a slope angle of 20.0°, and the man pulls upward with a force ***F*** whose direction makes an **angle of 30.0°** with the ramp. **(a) How large a force *F*** is necessary for the component F_x **parallel to the ramp to be 90.0 N**? . **(b) How large will the component F_y** perpendicular to the ramp be then?

$$F = \frac{F_X}{Cos(30^o)} = \frac{90}{Cos(30^o)}$$

$$F_Y = F\ Sin\ (30)$$

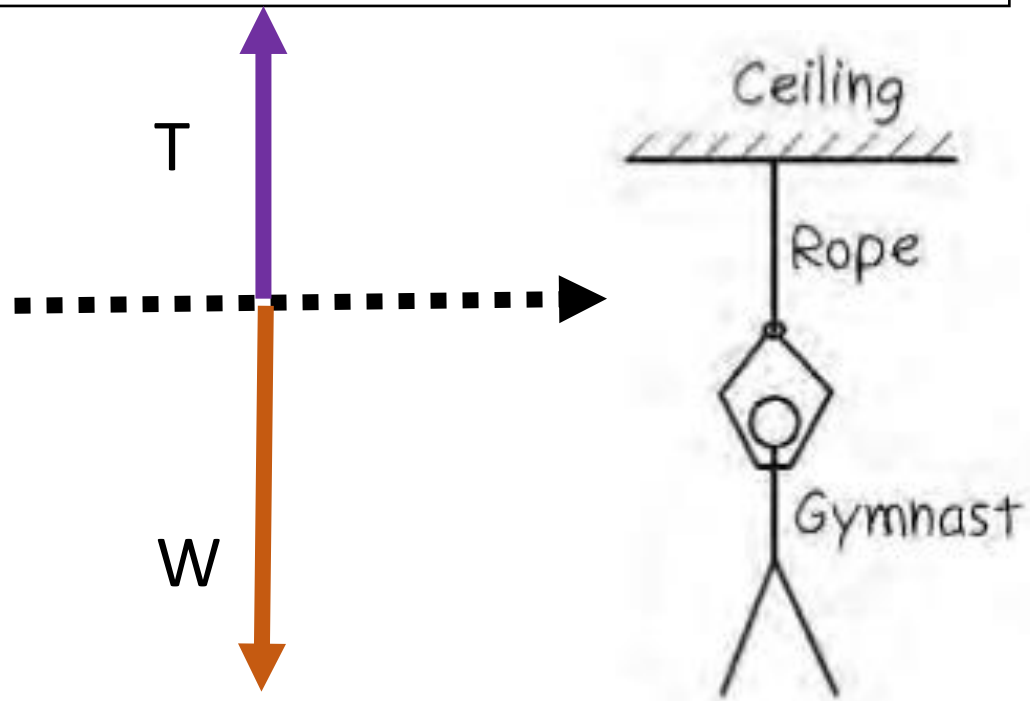


Example

A gymnast with mass $m_G = 50.0\ \text{kg}$ suspends herself from the lower end of a hanging rope of negligible mass. The upper end of the rope is attached to the gymnasium ceiling. **(a) What is the gymnast's weight? . (b) What force (magnitude and direction) does the rope exert on gymnast? (c) What is the tension at the top of the rope?**

$$W = m_G g = 50 * 9.8 = 490\ \text{N}$$

$$T = W = 490\ \text{N}$$

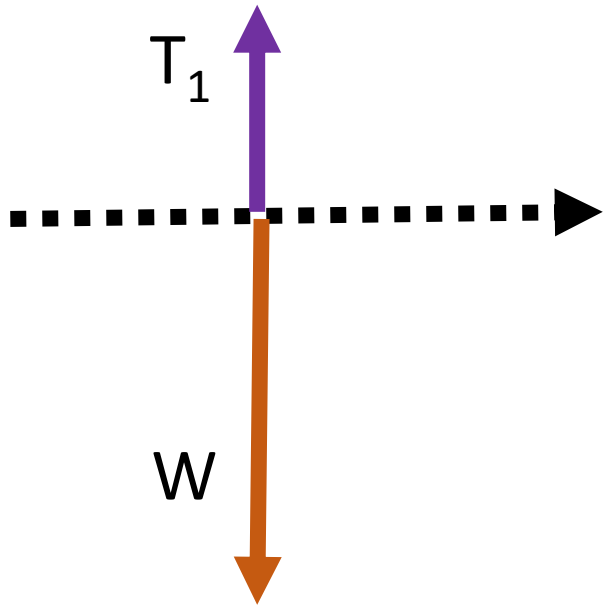


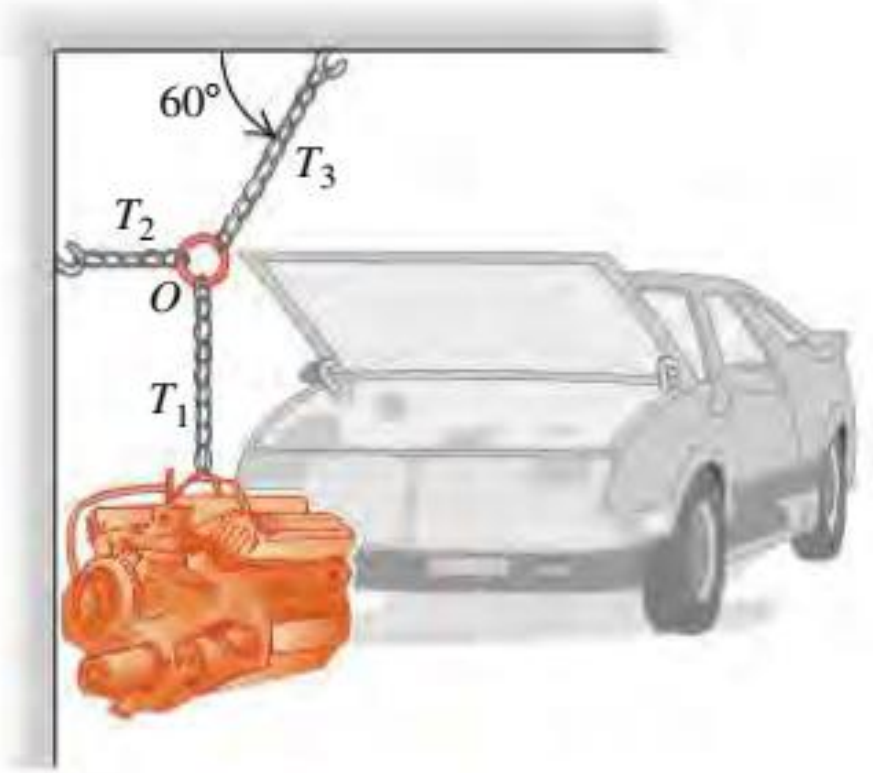
Example

A car engine with weight ***w*** hangs from a chain that is linked at ring *O* to two other chains, one fastened to the ceiling and the other to the wall. **Find expressions for the tension in each of the three chains in terms of *w***. The weights of the ring and chains are negligible compared with the weight of the engine.

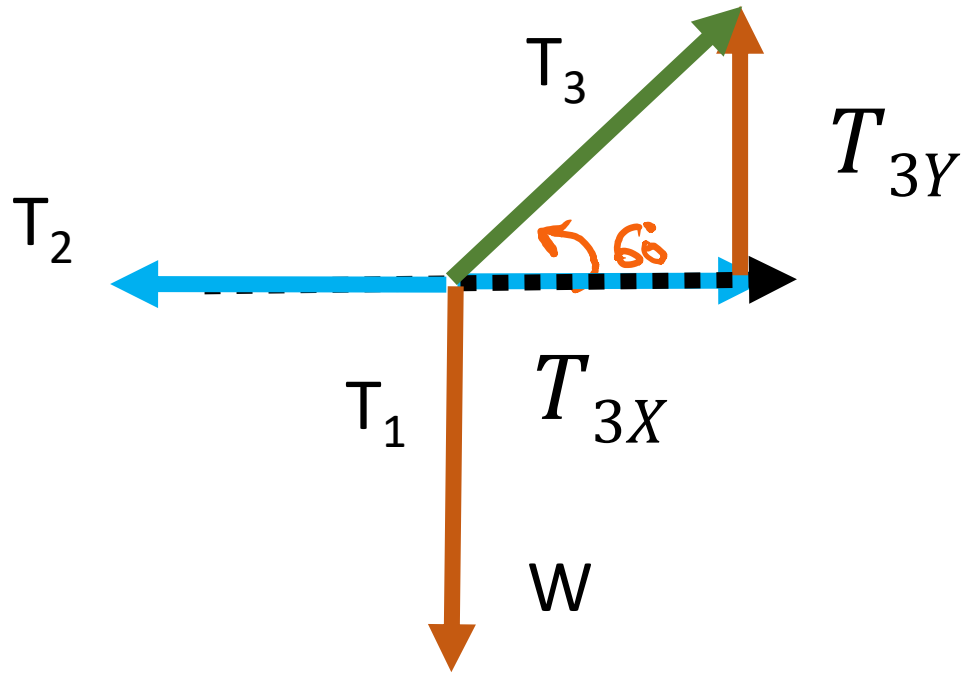
$$\sum F = 0$$

$$T_1 - W = 0$$



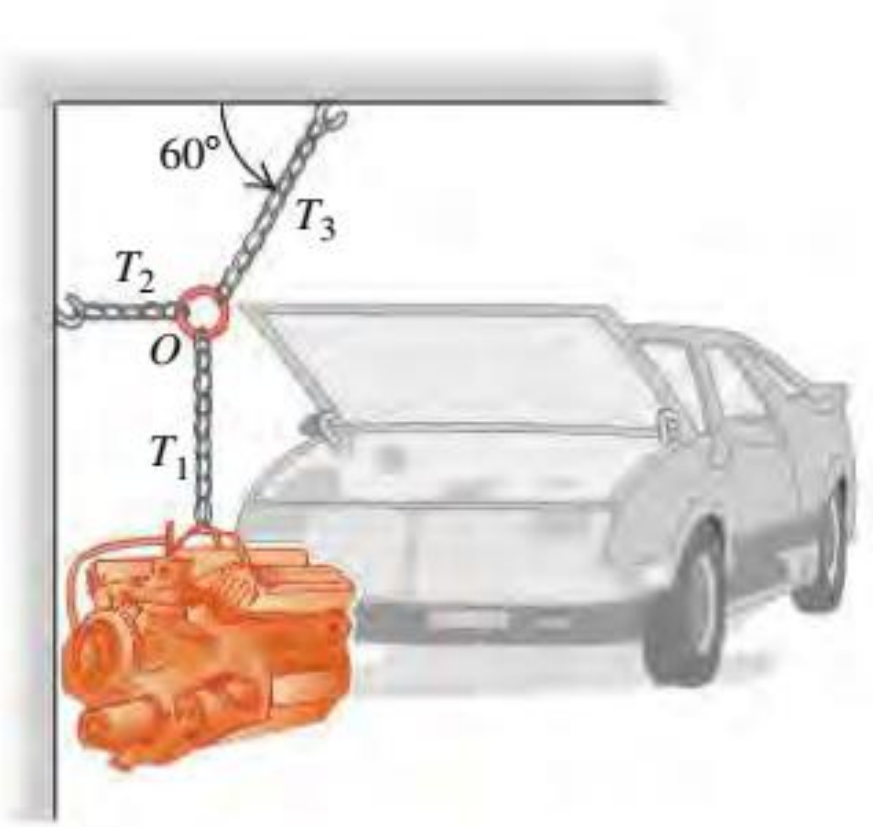


Example



T_{3Y}

T_{3X}



$$T_2 - T_{3X} = 0$$

$$T_1 + T_{3Y} = W$$

$$T_1 = W$$

Example

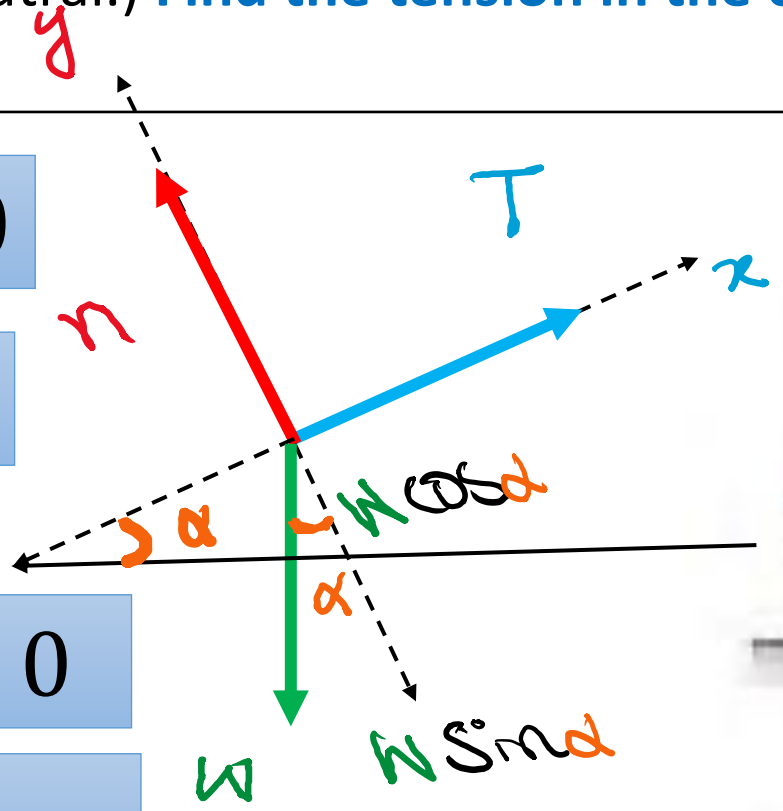
A car of **weight *w*** rests on a slanted ramp attached to a trailer. Only a cable running from the trailer to the car prevents the car from rolling off the ramp. (The car's brakes are off and its transmission is in neutral.) **Find the tension in the cable** and **the force that the ramp exerts on the car's tires**.

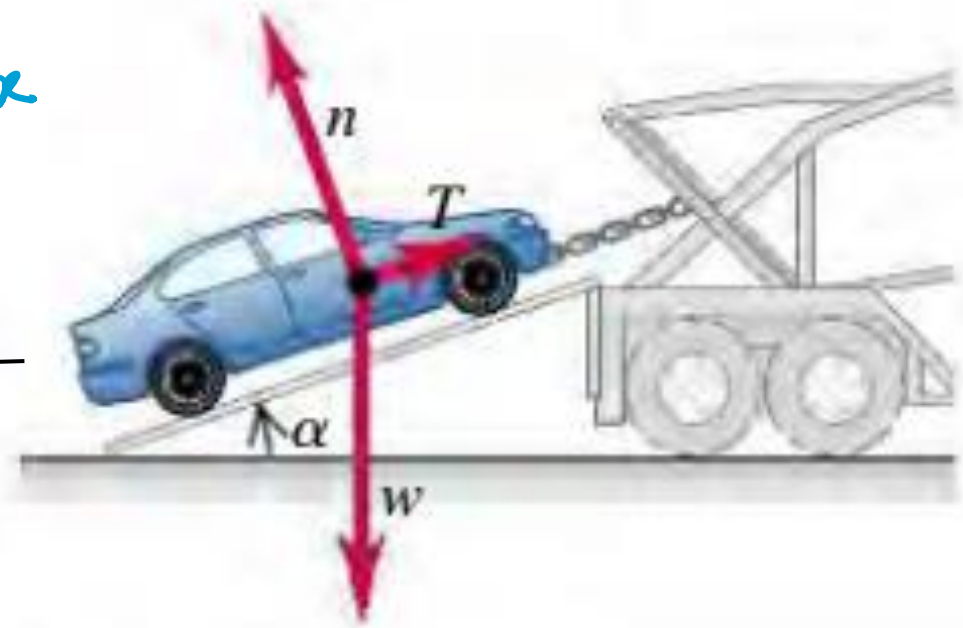
$$T - W\ Sin\ (\alpha) = 0$$

$$T = W Sin\ (\alpha)$$

$$n - W\ Cos\ (\alpha) = 0$$

$$n = W\ Cos\ (\alpha)$$





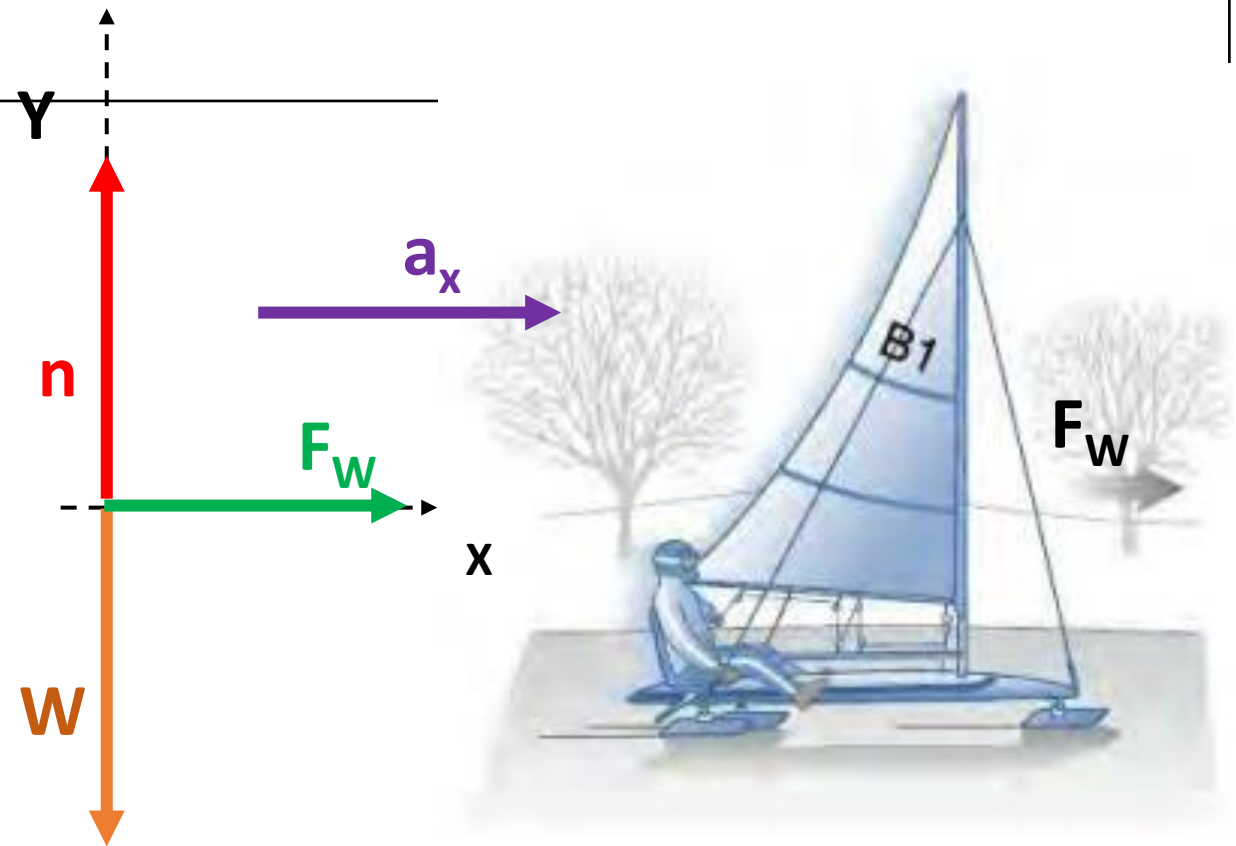
Example

An iceboat is at rest on a frictionless horizontal surface. Due to the blowing wind, 4.0 s after the iceboat is released, it is moving to the right at 6.0 m/s (about 22 km/h, or 13 mi/h). **What constant horizontal force F_W does the wind exert on the iceboat?** The combined mass of iceboat and rider is 200 kg.

$$\sum F_g = ma_y$$

$$n - W = 0$$

$$n = W$$



Example

An iceboat is at rest on a frictionless horizontal surface. Due to the blowing wind, 4.0 s after the iceboat is released, it is moving to the right at 6.0 m/s (about 22 km/h, or 13 mi/h). **What constant horizontal force F_W does the wind exert on the iceboat?** The combined mass of iceboat and rider is 200 kg.

$$\sum F_x = ma_x$$

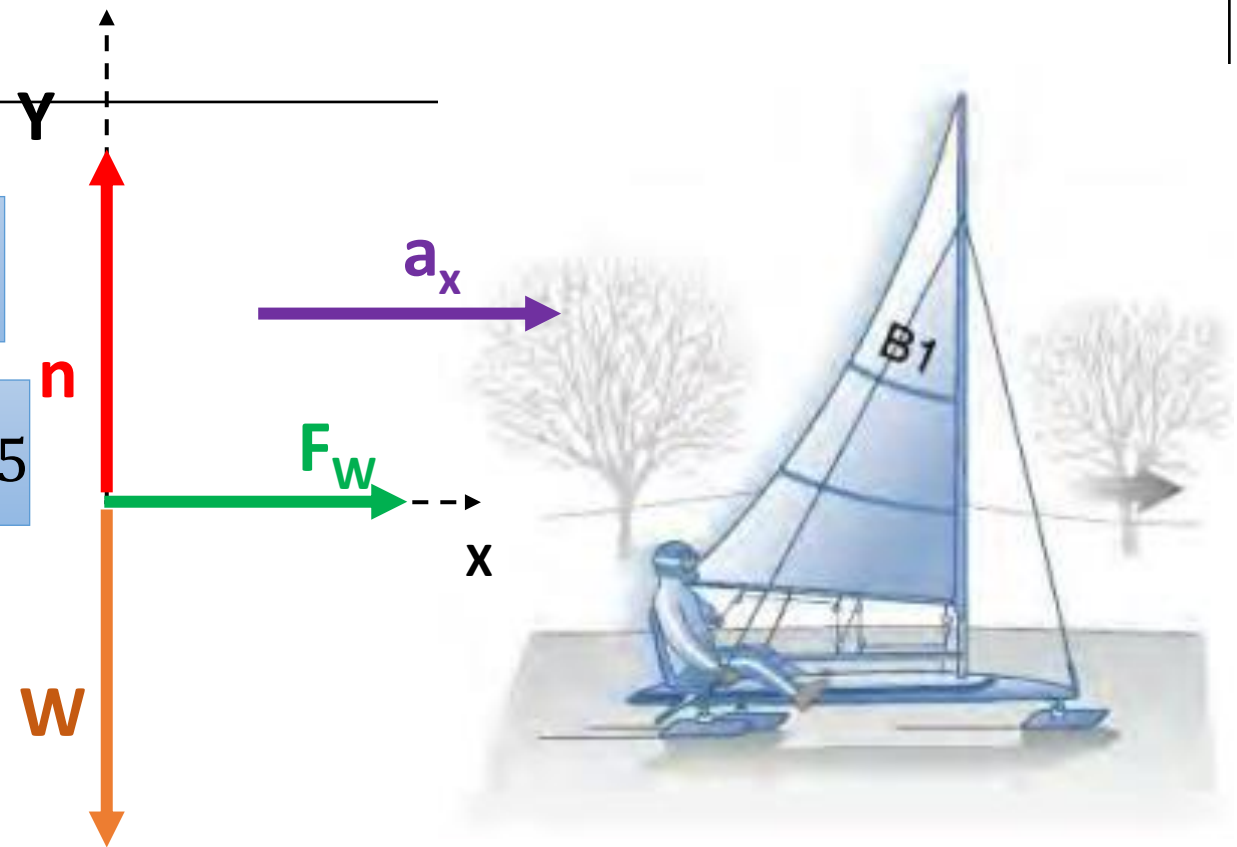
$$a_x = \frac{V_x - Vox}{t}$$

$$F_W = ma_x$$

$$a_x = \frac{6-0}{4} = 1.5$$

$$F_W = (200)a_x$$

$$F_W = (200)(1.5) = 300N$$



Example

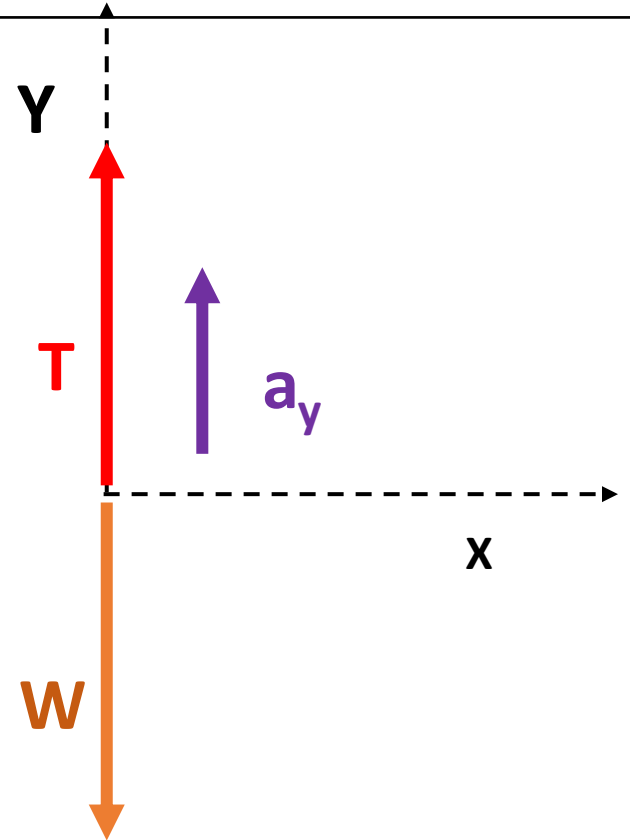
An elevator and its load have a combined **mass of 800 kg.** The elevator is initially moving downward at 10.0 m/s; it slows to a stop with constant acceleration in a distance of 25.0 m. **What is the tension *T* in the supporting cable** while the elevator is being brought to rest?

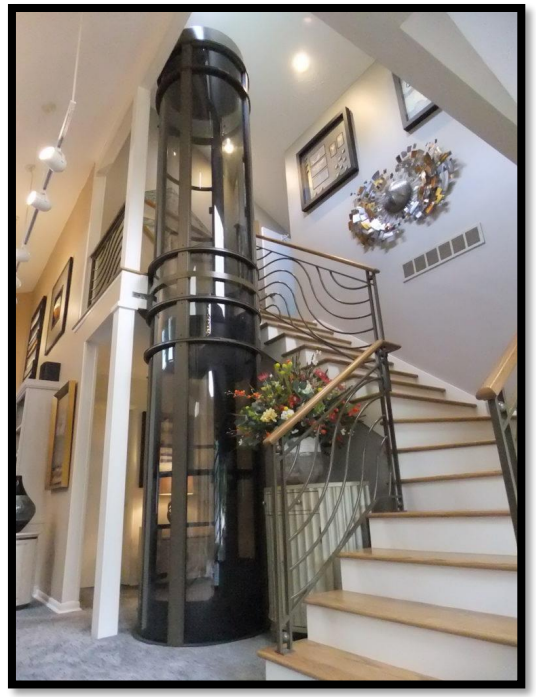
$$\sum F_g = ma_y$$

$$T - W = ma_y$$

$$\mathrm{T} = \mathrm{W} + ma_y$$

$$\mathrm{T} = \mathrm{mg} + ma_y$$





Example

An elevator and its load have a combined **mass of 800 kg.** The elevator is initially moving downward at 10.0 m/s; it slows to a stop with constant acceleration in a distance of 25.0 m. **What is the tension *T* in the supporting cable** while the elevator is being brought to rest?

$$T = mg + ma_y$$

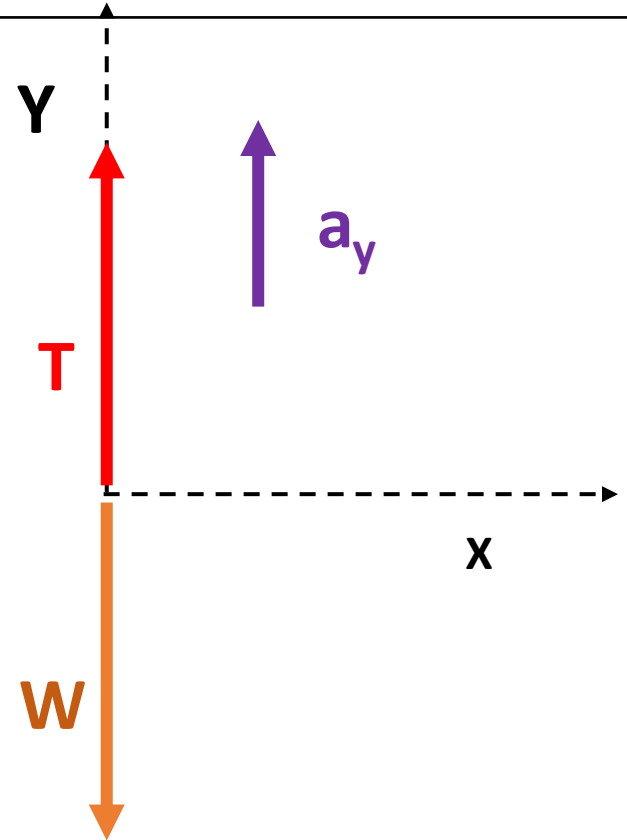
$$T = m(g + a_y)$$

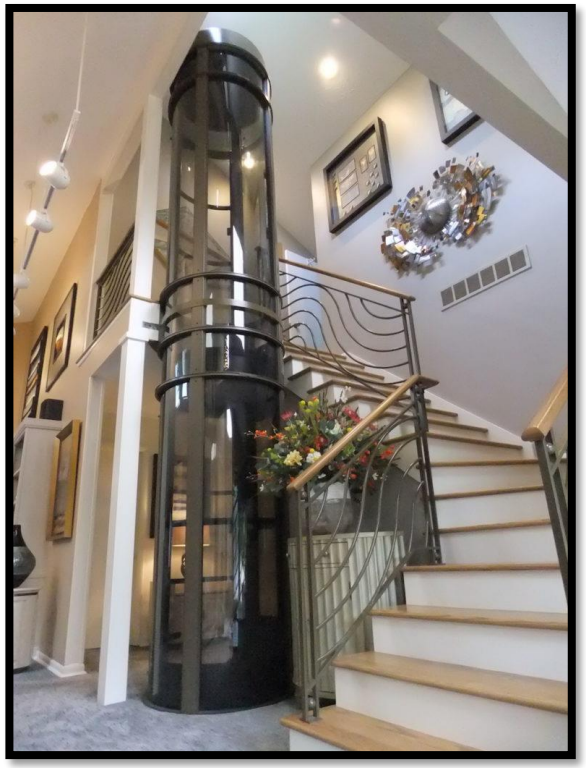
$$T = 800(9.8 + a_y)$$

$$v_y^2 = v_{0y}^2 + 2a_y(y - y_0)$$

$$(10)^2 = (0)^2 + 2a_y(25)$$

$$a_y = 2\ m/s^2 \qquad T = 9440\ N$$





Example

An elevator is initially moving downward at 10.0 m/s; it slows to a stop with constant acceleration in a distance of 25.0 m. A 50 kg person stands on a bathroom scale while riding in the elevator. What is the reading on the scale?

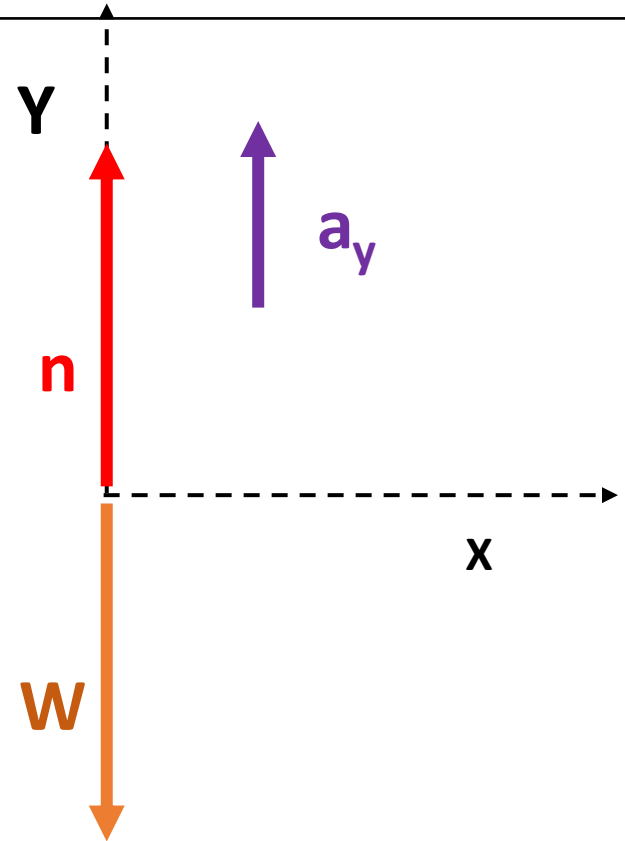
$$\sum F_y = ma_y$$

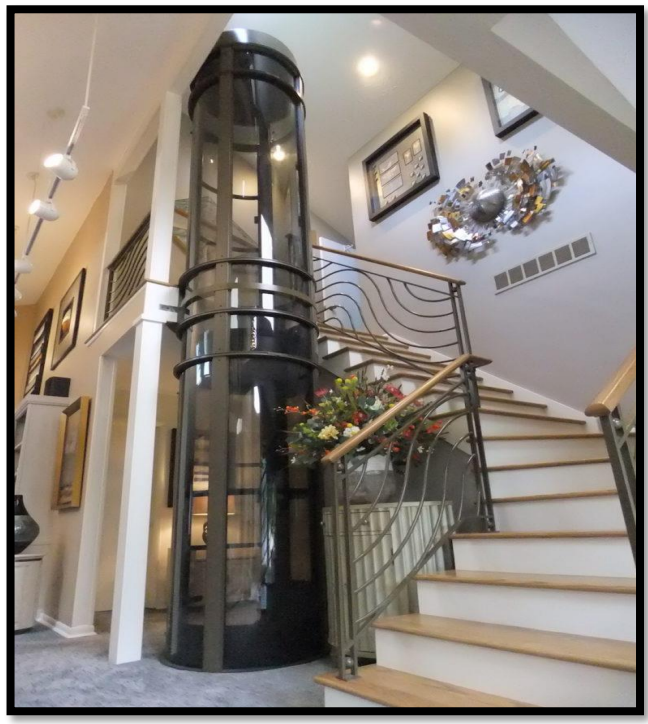
$$n - W = may$$

$$\mathrm{n} = \mathrm{mg} + ma_y$$

$$\mathrm{n} = \mathrm{m}(\mathrm{g} + a_y)$$

$$\mathrm{n} = 50(9.8 + a_y)$$





Example

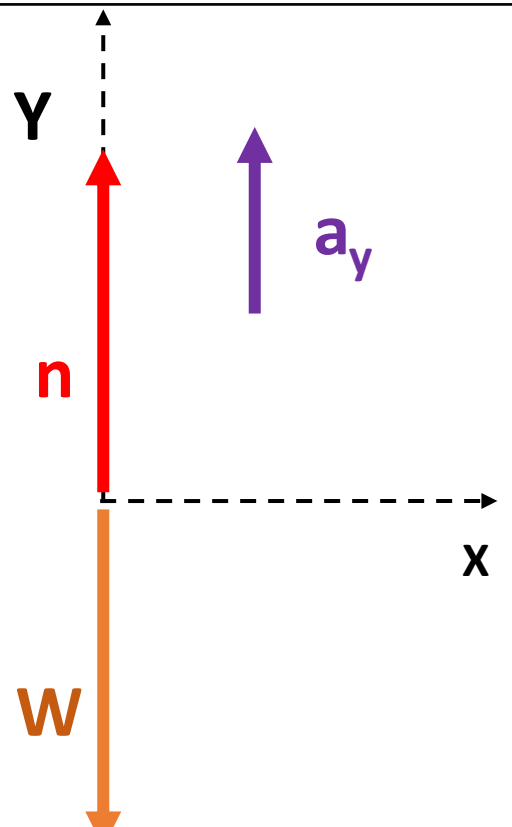
An elevator is initially moving downward at 10.0 m/s; it slows to a stop with constant acceleration in a distance of 25.0 m. A 50.0-kg person stands on a bathroom scale while riding in the elevator. What is the reading on the scale?

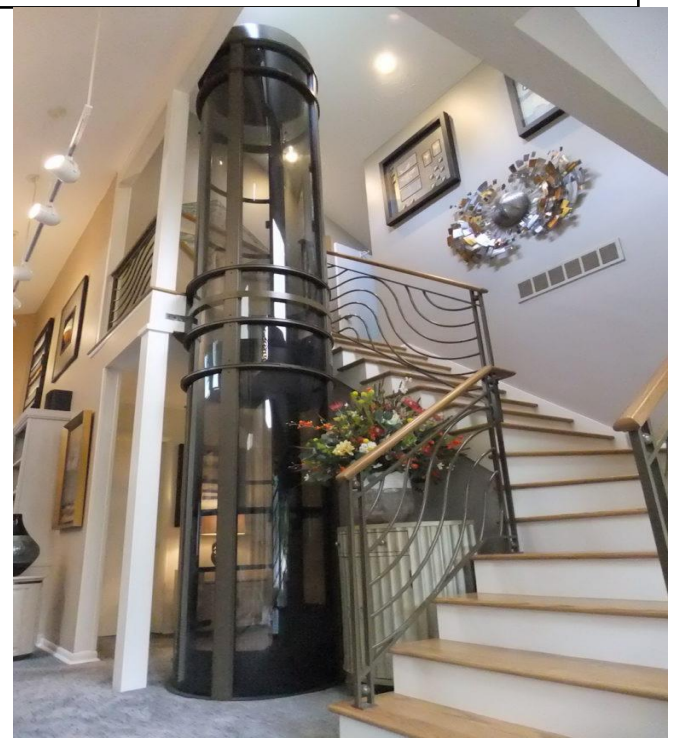
$$v_y^2 = v_{0y}^2 + 2a_y(y - y_0)$$

- u = 10.0 m/s
- v = 0 m/s
- s = 25.0 m
- a = ?

$$a_y = -2.0\ \text{m/s}^2$$

$$n = 50(9.8 + a_y)$$





Forces (Contact Forces)

Forces that occur when two objects are physically touching each other.

Force Type	Description	Real-World Example
Frictional Force	Opposes relative motion between surfaces in contact	Walking, car brakes, sliding a book
Normal Force	Perpendicular force exerted by a surface on an object	Book resting on a table, person standing on floor
Tension Force	Pulling force through a string, rope, or cable	Towing a car, elevator cable, hanging sign
Applied Force	Direct push or pull on an object	Pushing a shopping cart, kicking a ball
Air Resistance	Frictional force from air molecules on moving objects	Parachute, skydiver falling
Spring Force	Restoring force from compressed/stretched spring	Car suspension, mattress springs
Buoyant Force	Upward force from fluid on submerged objects	Boat floating, helium balloon rising

Forces (Non-Contact Forces)

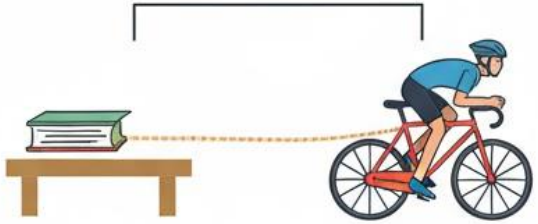
Forces that occur when two objects are physically touching each other.

Force Type	Description	Real-World Example
Gravitational Force	Attractive force between masses	Planets orbiting sun, objects falling
Magnetic Force	Force between magnetic poles or moving charges	Compass needle, MRI machines
Electric Force	Force between charged particles	Static electricity, lightning
Nuclear Forces	Forces within atomic nuclei	Holding protons/neutrons together

CONTACT FORCES
(Touching Required)



Friction
(sliding, rolling)



Tension
Reistance
(drag)

Air Resistance
(ropes, cables)

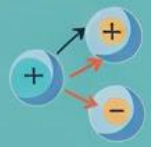


Buayancy
(fluids)

NON-CONTACT FORCES
(Act at Distance)



Gravational
(weight, orbits)

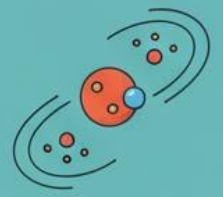




Electomgenetic
(magnets, charges)

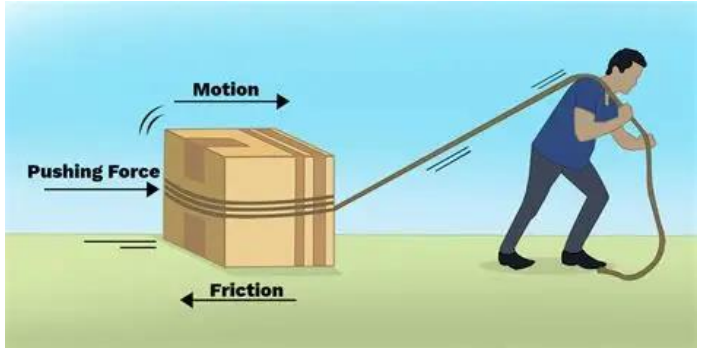


Strong Nuclear
(atomic nuclei)



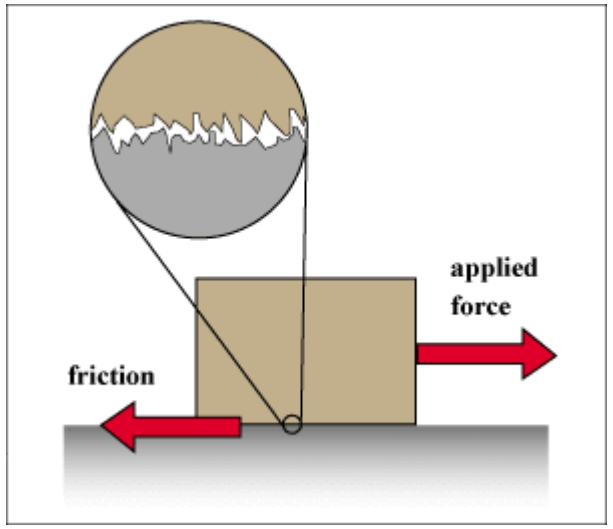
Weak Nuclear
(radicative decay)

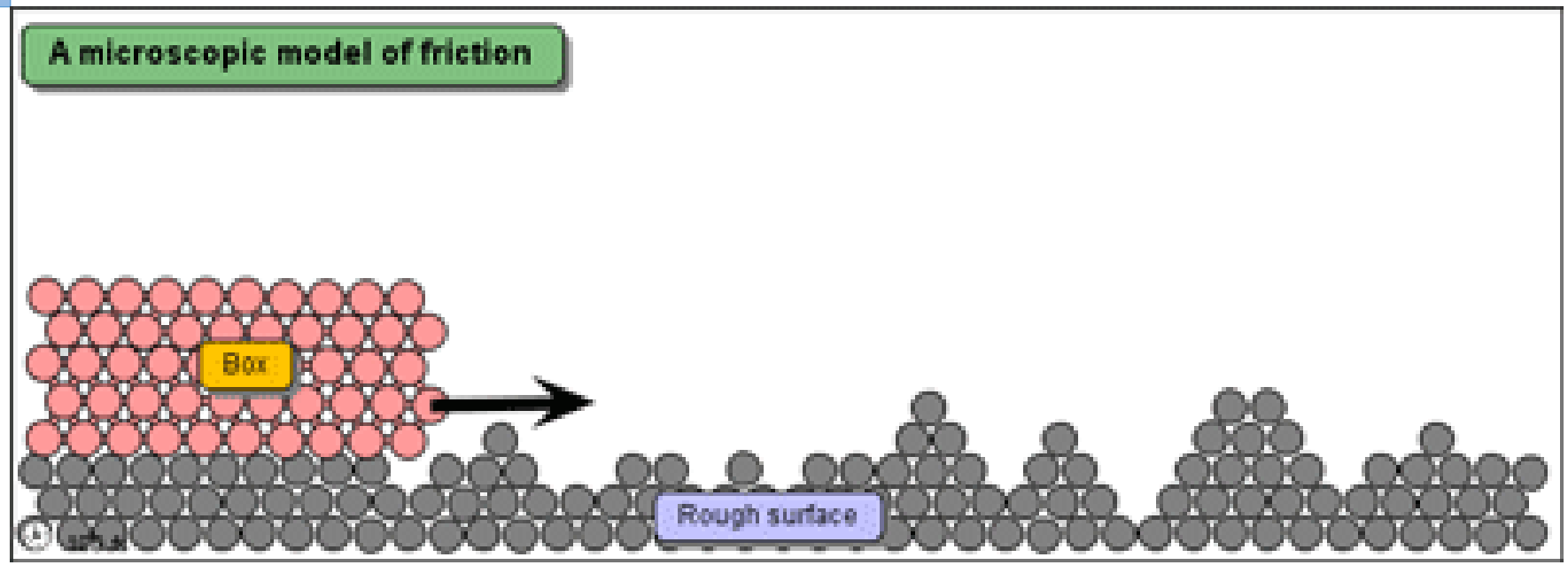
Friction



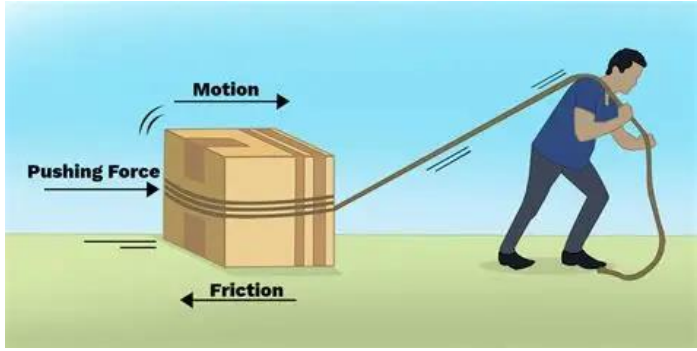
$$f_k \propto n$$

$$f_k = \mu_k n$$



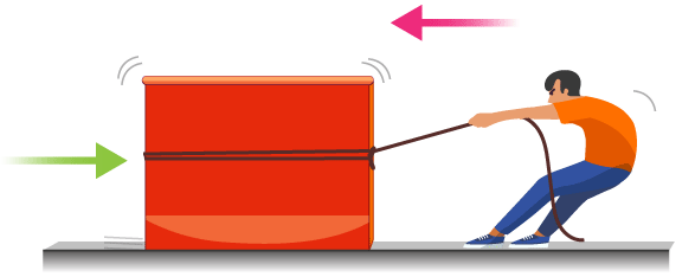


Friction

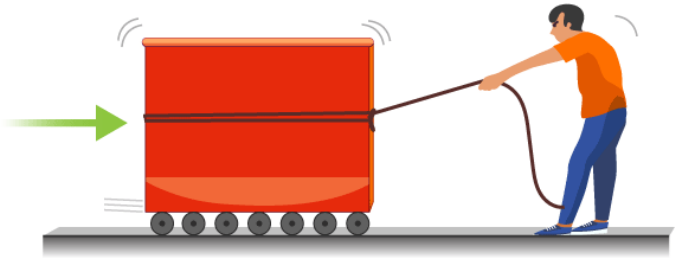


$$f_k \propto n$$

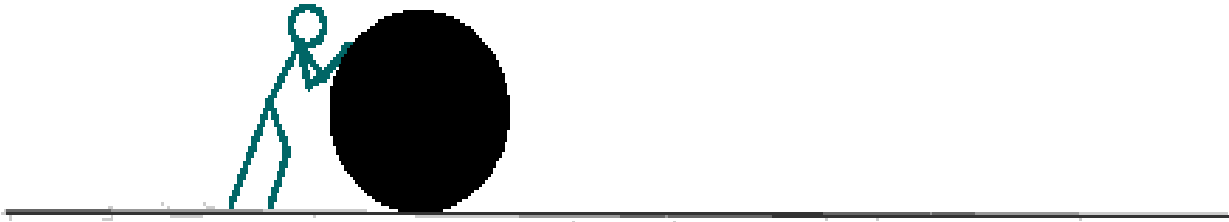
$$f_k = \mu_k n$$



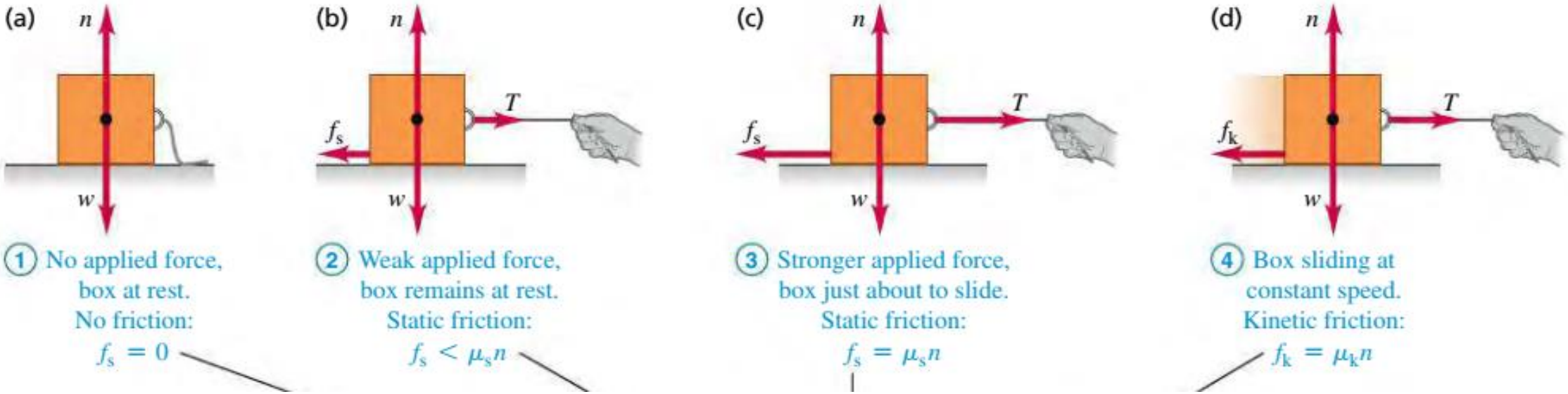
Sliding Friction



Rolling Friction





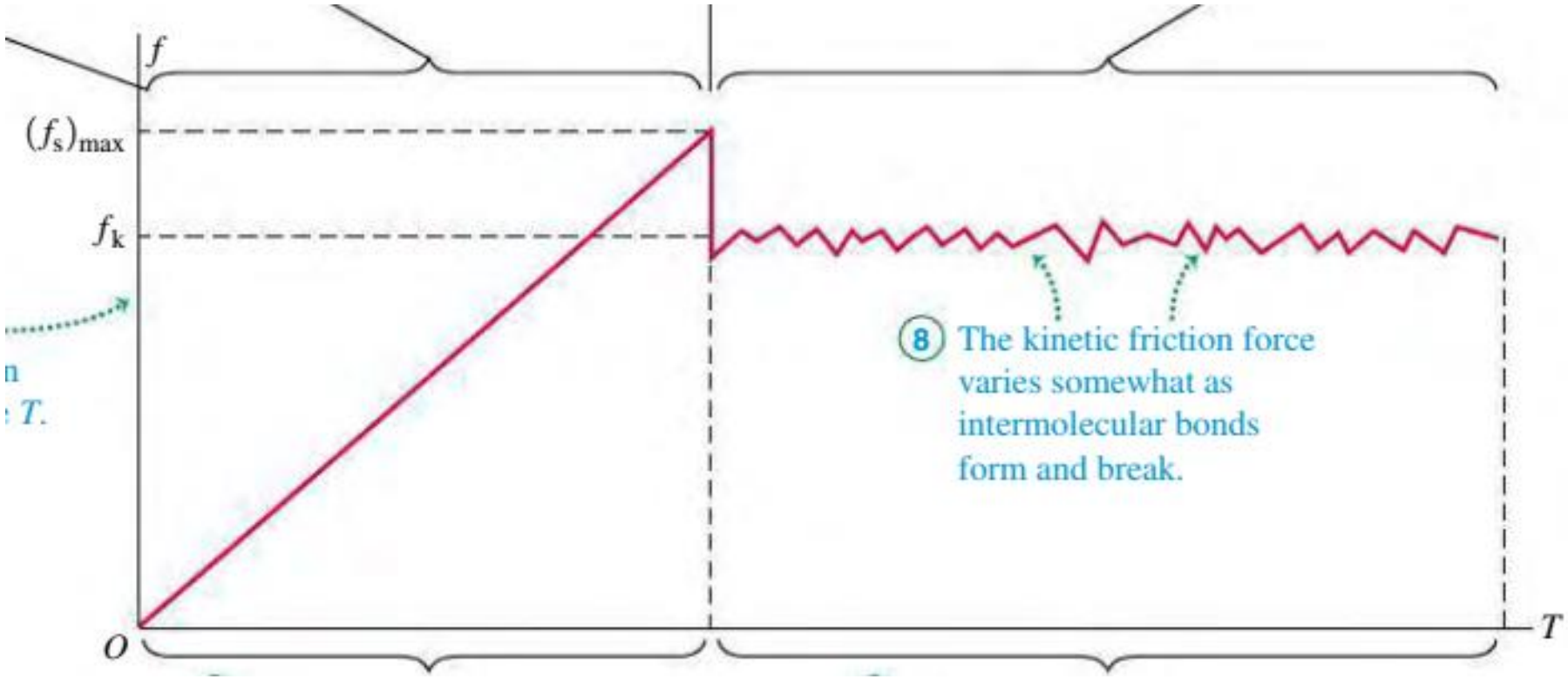


(1) No applied force, box at rest. No friction: $f_s = 0$

(2) Weak applied force, box remains at rest. Static friction: $f_s < \mu_s n$

(3) Stronger applied force, box just about to slide. Static friction: $f_s = \mu_s n$

(4) Box sliding at constant speed. Kinetic friction: $f_k = \mu_k n$

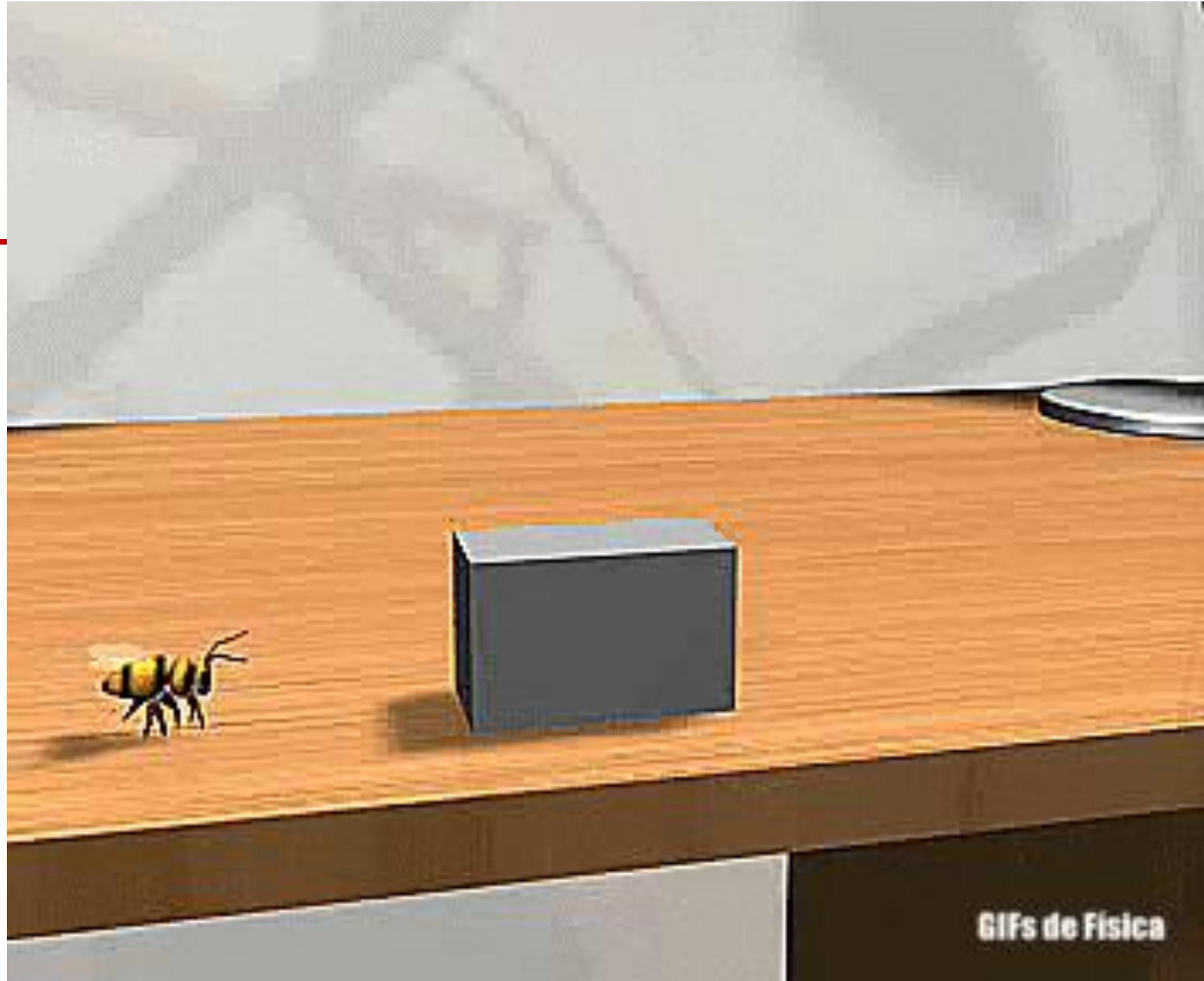


(8) The kinetic friction force varies somewhat as intermolecular bonds form and break.

Friction

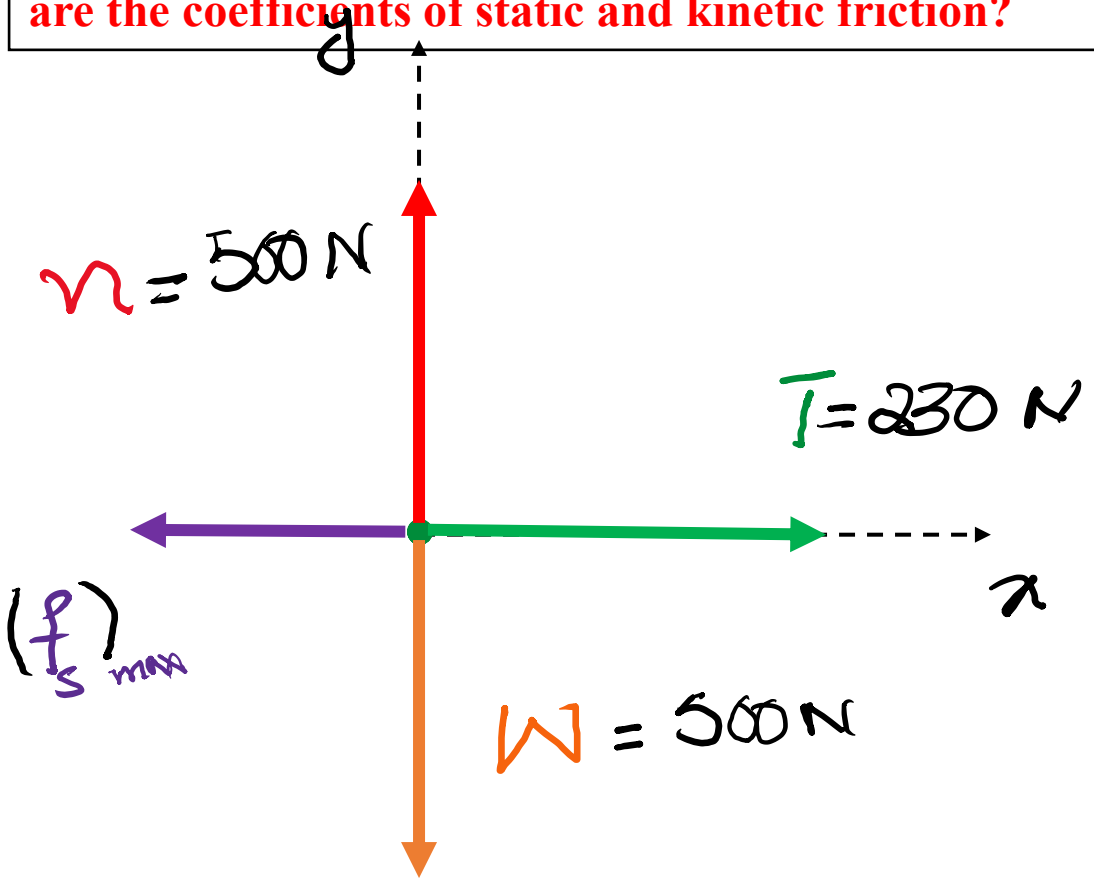
$$f_k = \mu_k n$$

$$f_S = \mu_S n$$



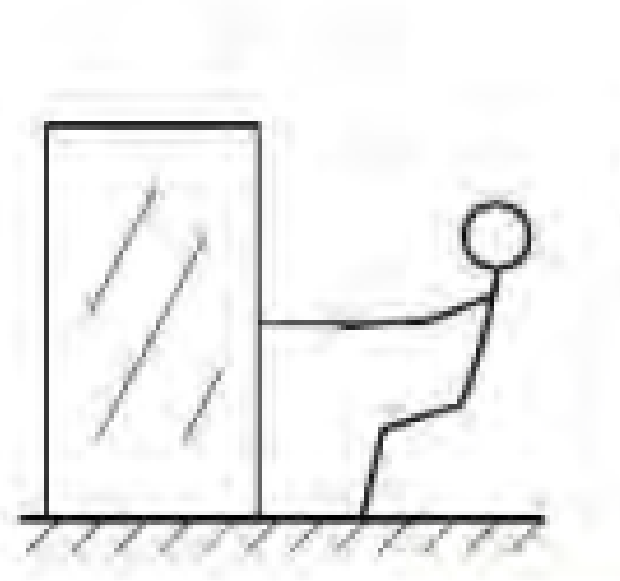
Friction

You want to move a **500-N crate** across a level floor. To start the crate moving, you have to pull with a **230-N horizontal force**. Once the crate starts to move, you can keep it moving at constant velocity **with only 200 N**. **What are the coefficients of static and kinetic friction?**



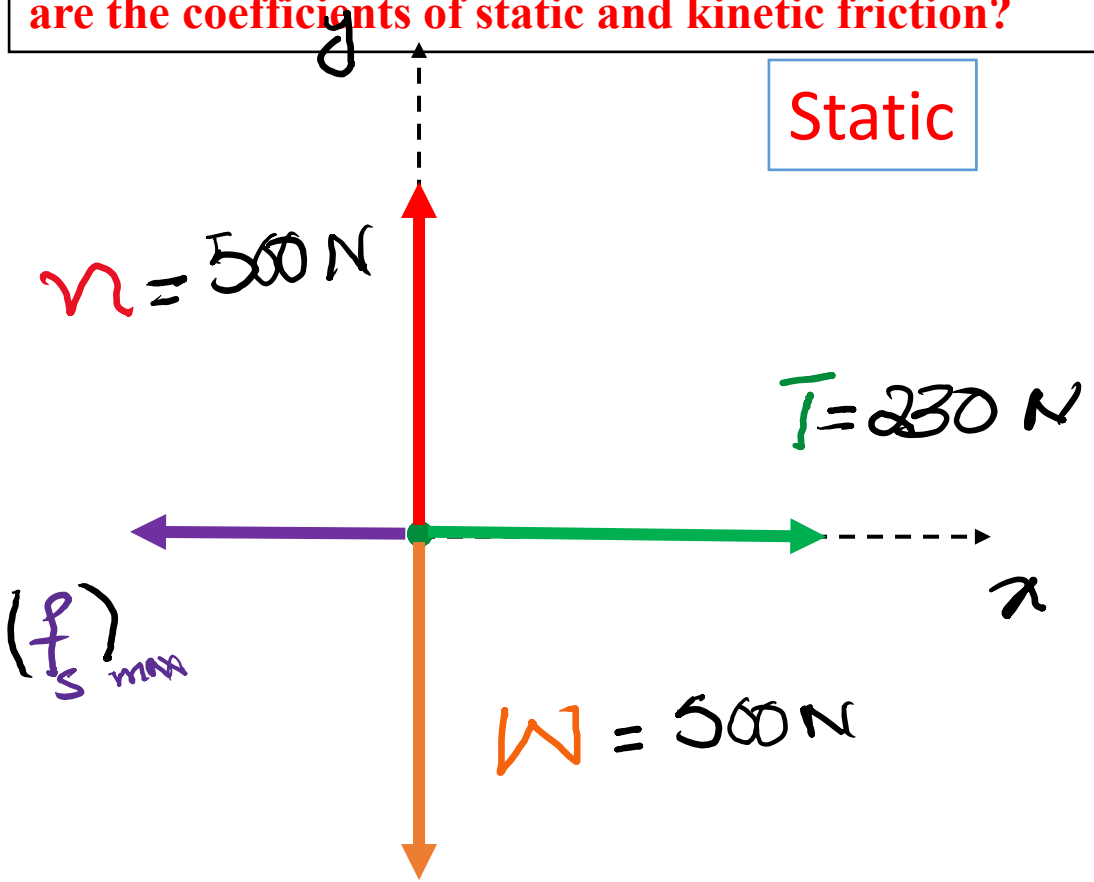
$$n - W = 0$$

$$n = W$$



Friction

You want to move a **500-N crate** across a level floor. To start the crate moving, you have to pull with a **230-N horizontal force**. Once the crate starts to move, you can keep it moving at constant velocity **with only 200 N**. **What are the coefficients of static and kinetic friction?**



Static

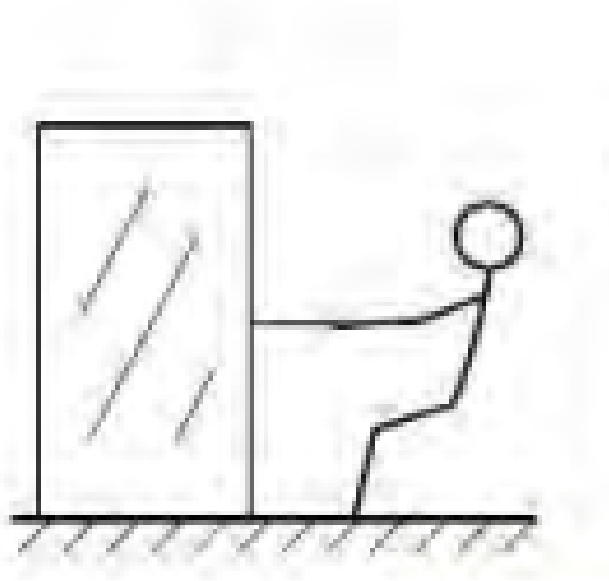
$$T - f_S = 0$$

$$f_S = T$$

$$f_S = T = 230\ N$$

$$f_s = \mu_S n$$

$$\mu_S = \frac{f_S}{n} = \frac{230}{500}$$



Friction

You want to move a **500-N crate** across a level floor. To start the crate moving, you have to pull with a **230-N horizontal force**. Once the crate starts to move, you can keep it moving at constant velocity **with only 200 N**. **What are the coefficients of static and kinetic friction?**

$$T - f_k = 0$$

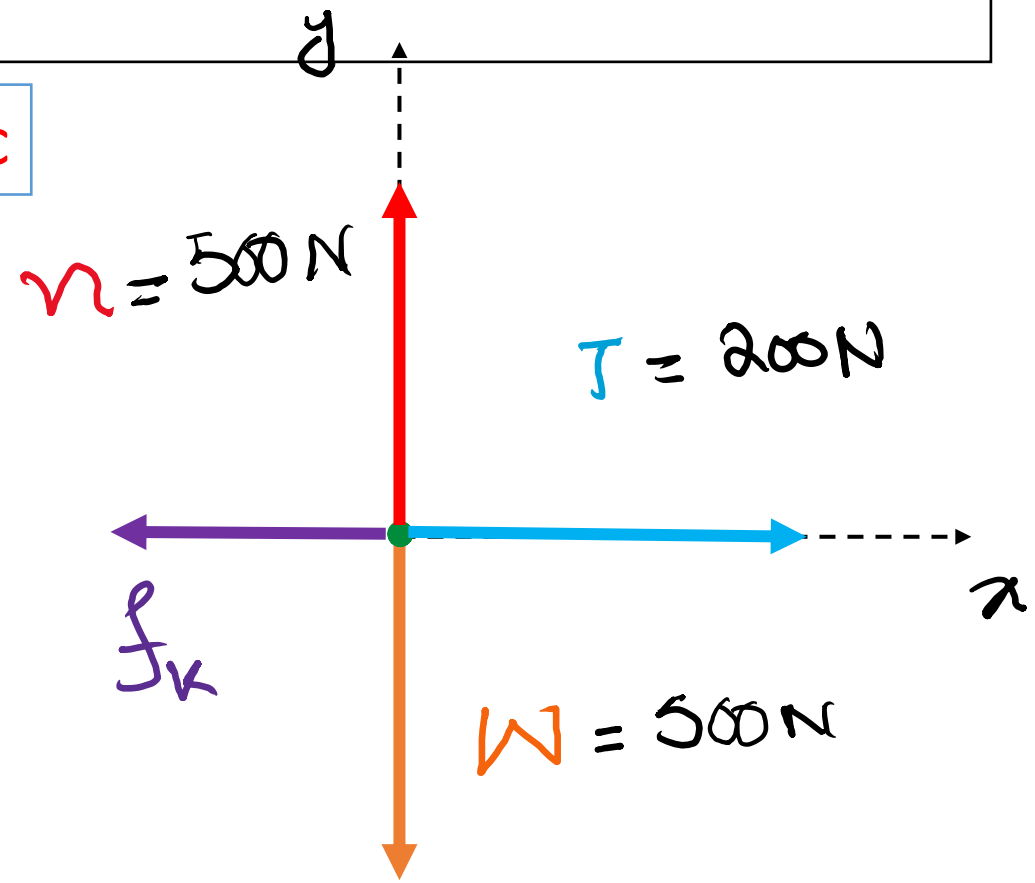
$$f_k = T$$

$$f_k = T = 200\ N$$

$$f_k = \mu_k n$$

$$\mu_k = \frac{f_k}{n} = \frac{200}{500}$$

Kinetic



Friction

You want to move a **500-N crate** across a level floor. suppose you move the crate by pulling upward on the rope at an **angle of 30°** above the horizontal. **How hard must you pull** to keep it moving with constant velocity? Assume that $\boldsymbol{\mu_k = 0.40}$

$$T_x - f_k = 0$$

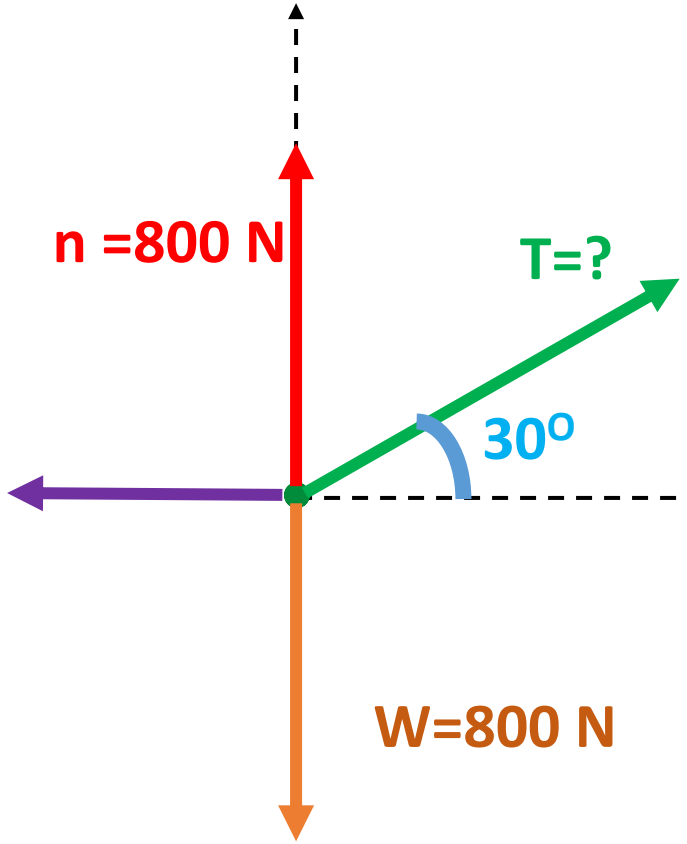
$$f_k = \mu_k n$$

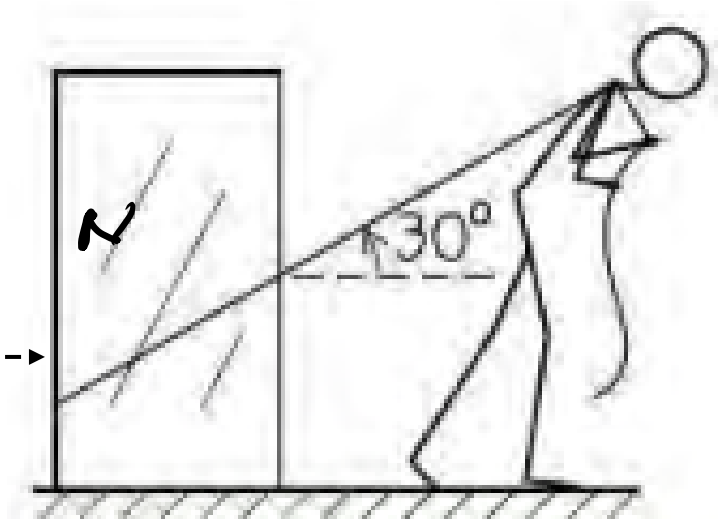
$$f_k = \mu_k n = T\cos(30^o)$$

$$T_y + n - W = 0$$

$$T\,\mathrm{Sin}(30^o) + n - W = 0$$

$$T\,\mathrm{Sin}(30^o) + n - 500 = 0$$





Friction

A group of students (**total weight** w) slides down a snow-covered hill that slopes at a constant angle α. There is a nonzero coefficient of kinetic friction μ_k. The slope has just the right angle to make the toboggan slide with constant velocity. **Find this angle in terms of w and μ_k.**

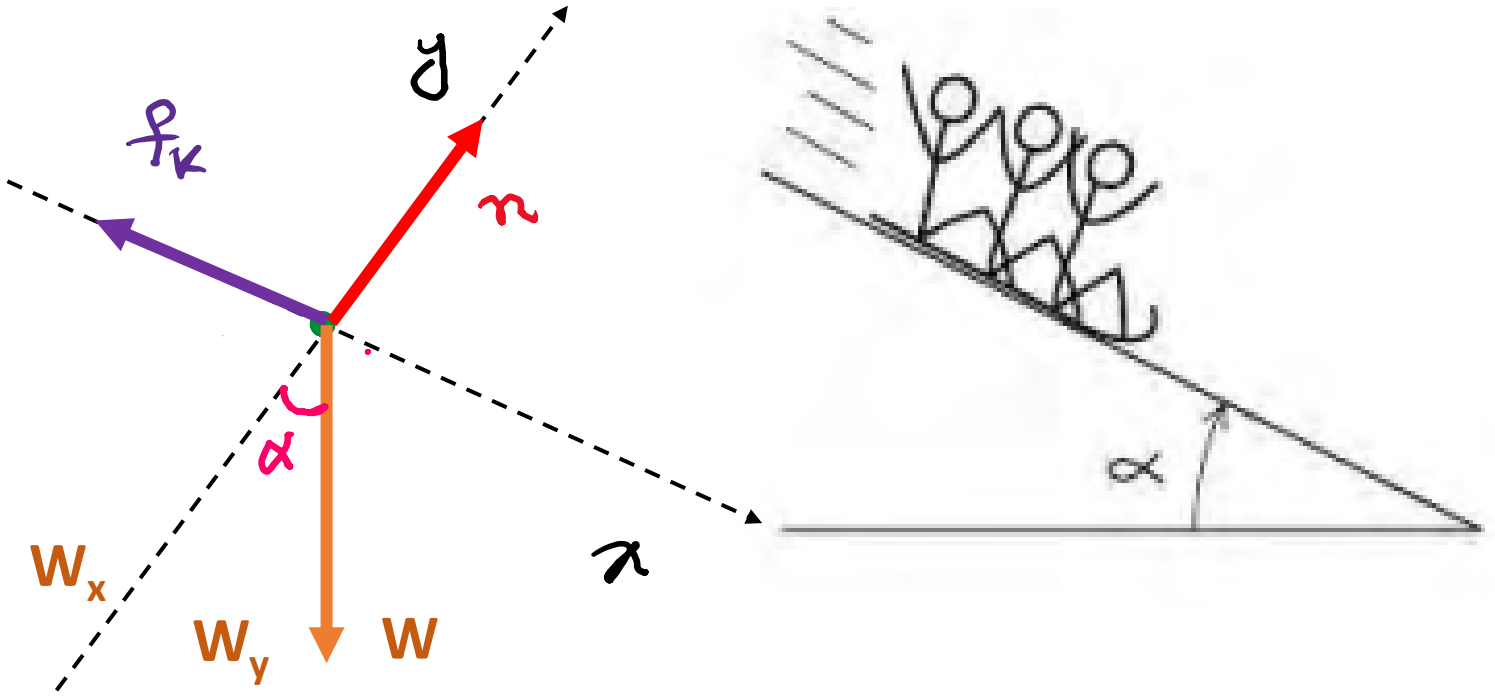
$$W_y - f_k = 0$$

$$f_k = \mu_k n$$

$$n - W_y = 0$$

μ_k=?

W=?



Example

A group of students (**total weight** w) slides down a snow-covered hill that slopes at a constant angle α. There is a nonzero coefficient of kinetic friction μ_k. The slope has just the right angle to make the object (toboggan) slide with constant velocity. **Find this angle in terms of w and μ_k.**

$$W_y - f_k = 0$$

$$f_k = \mu_k n$$

$$W_y - \mu_k n = 0$$

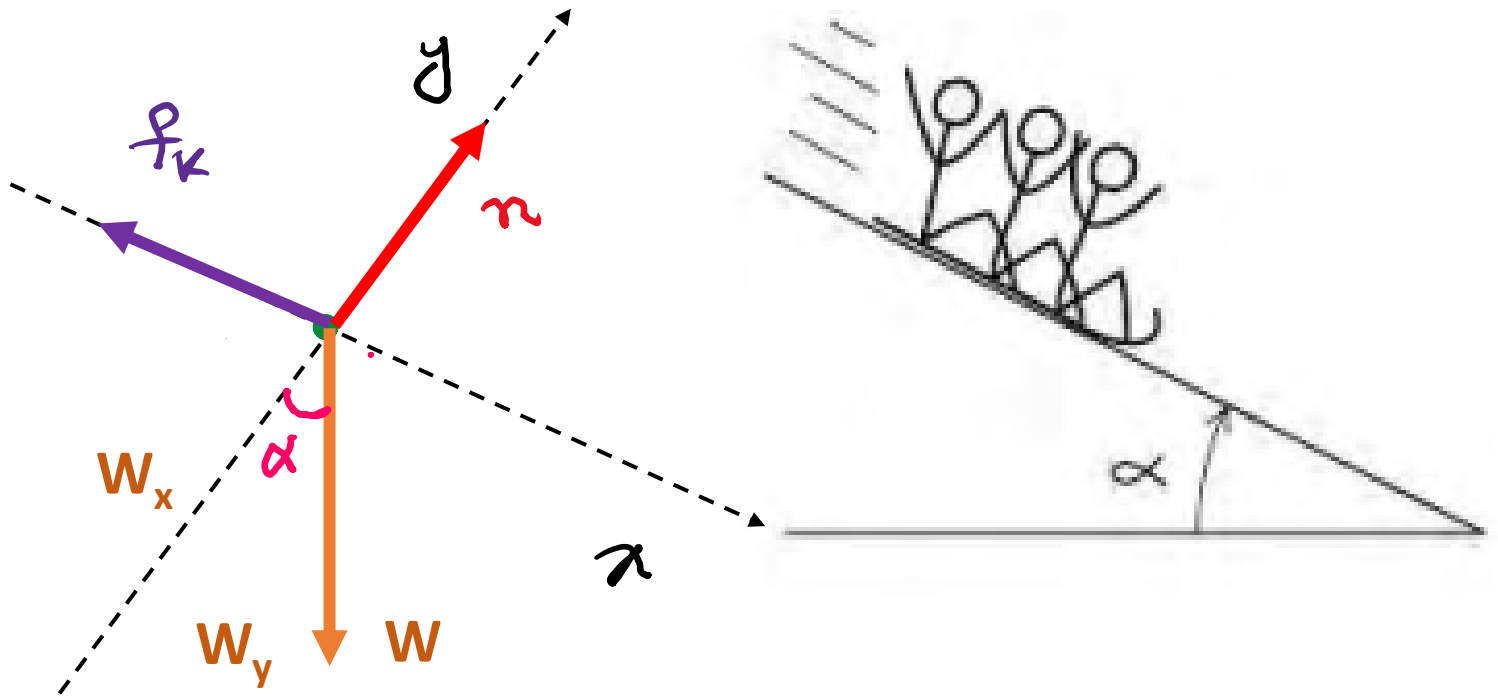
$$W\ Sin\ (\alpha) - \mu_k n = 0$$

$$n - W_x = 0$$

$$n = W\ Cos\ (\alpha)$$

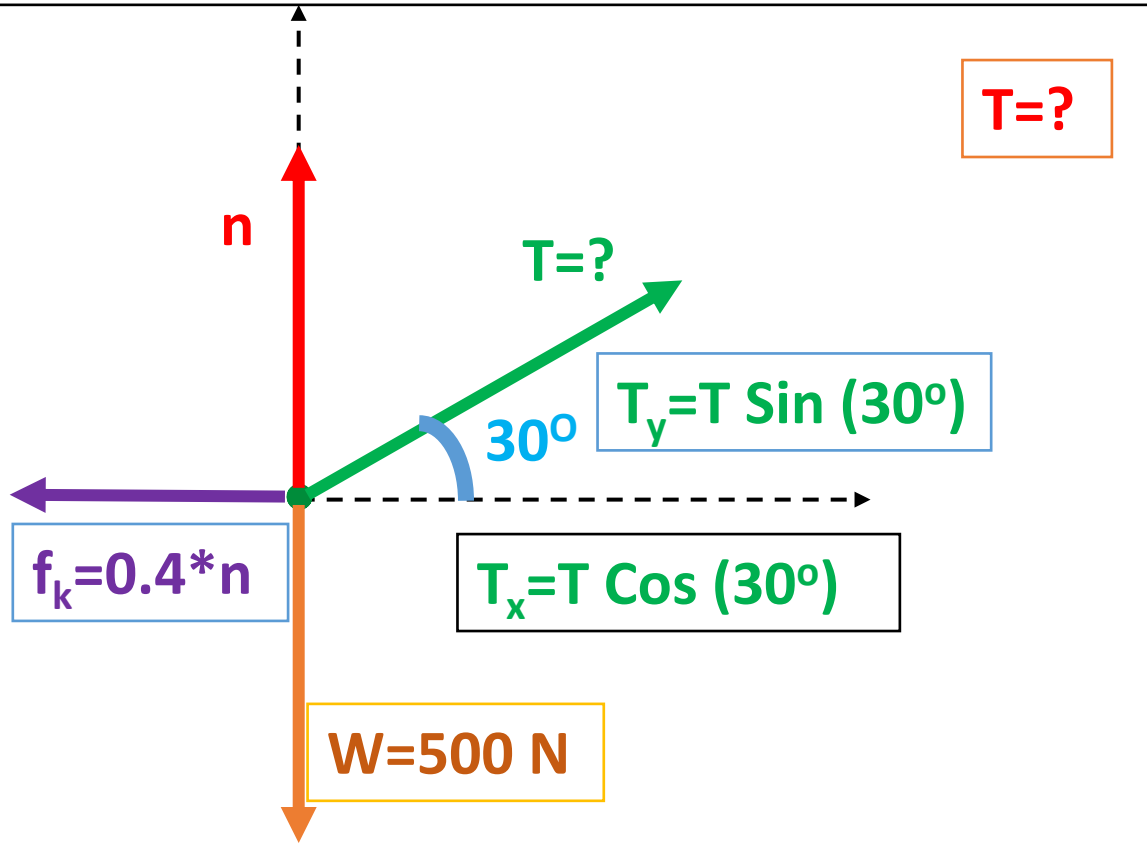
μ_k=?

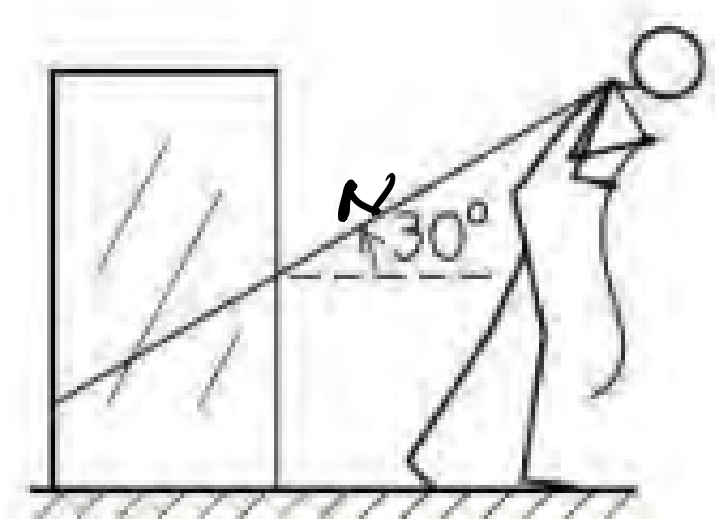
W=?



Example

You want to move a **500-N crate** across a level floor. suppose you move the crate by pulling upward on the rope at an **angle of 30°** above the horizontal. **How hard must you pull** to keep it moving with constant velocity? Assume that $\mu_k = 0.40$





Example

You want to move a **500-N crate** across a level floor. suppose you move the crate by pulling upward on the rope at an **angle of 30°** above the horizontal. **How hard must you pull** to keep it moving with constant velocity? Assume that $\mu_k = 0.40$

$$T_x - f_k = 0$$

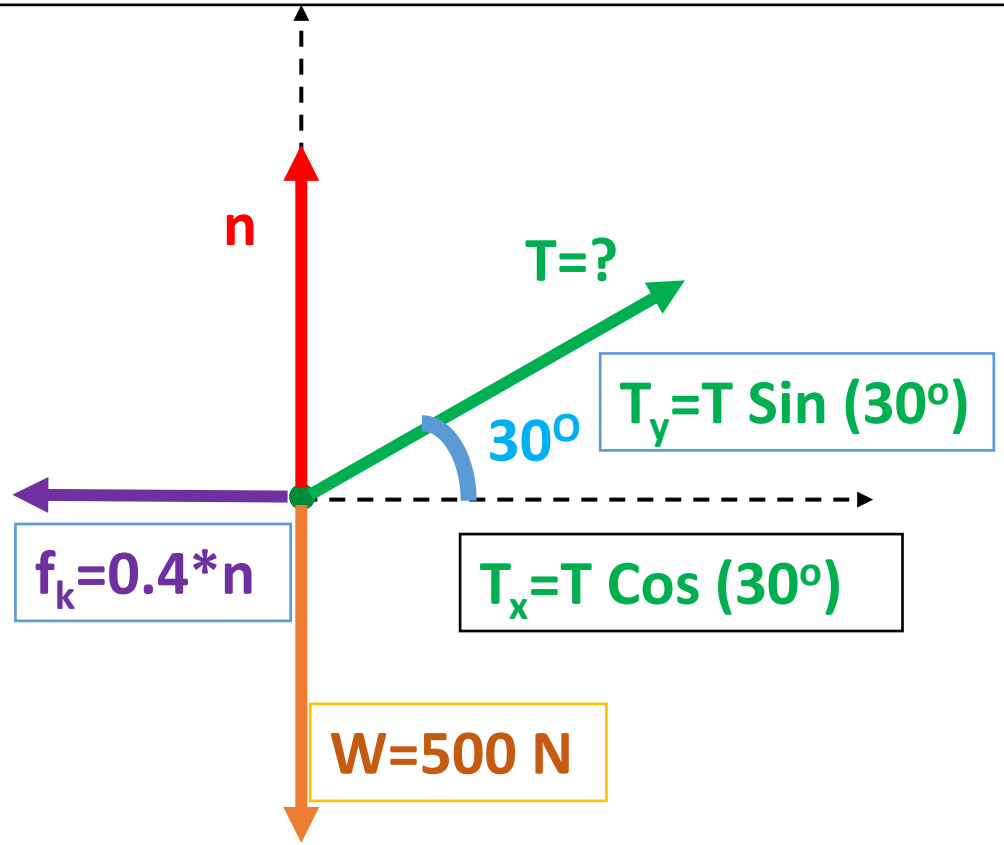
$$T_x - fk = \mu_k n$$

$$(0.4)n = T\cos(30^o)$$

$$T_y + n - W = 0$$

$$T\ \mathrm{Sin}(30^o) + n - W = 0$$

$$T\ \mathrm{Sin}(30^o) + n - 500 = 0$$



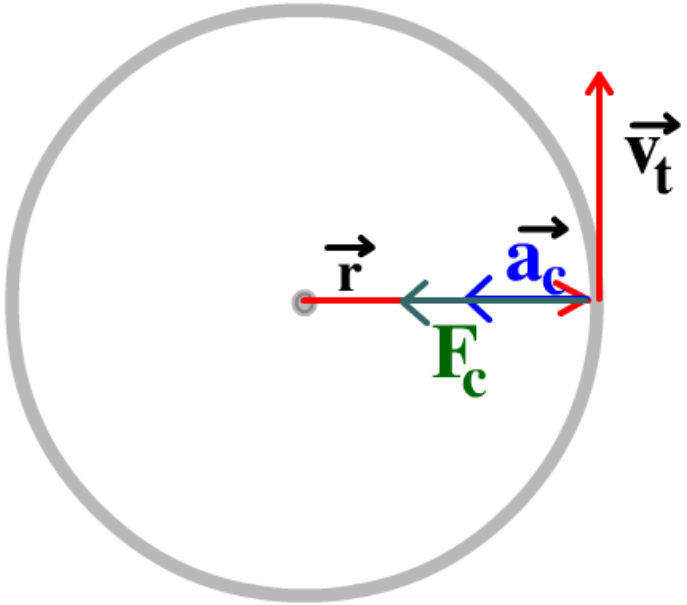
Dynamics of Circular Motion

$$a_{rad} = \frac{V^2}{R} = \frac{4\pi^2 R}{T^2}$$

$$F = ma_{rad}$$

$$F = m\frac{V^2}{R}$$

$$F = m\frac{4\pi^2 R}{T^2}$$



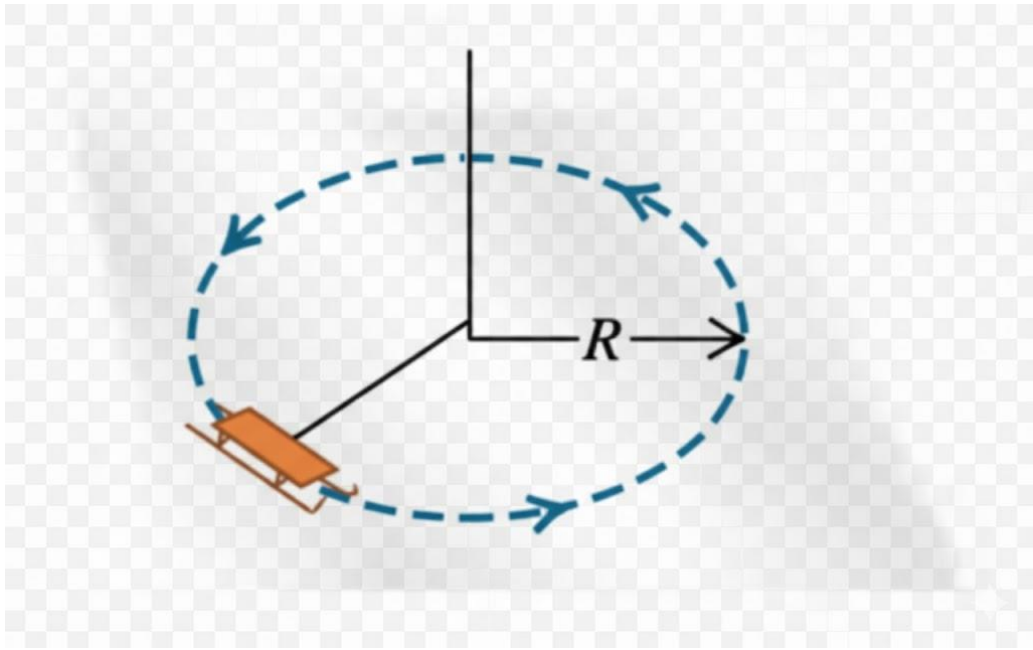
Example

A toy with a mass of **25.0 kg** rests on a horizontal sheet of essentially frictionless ice. It is attached by a **5.00-m rope** to a post set in the ice. Once given a push, the toy revolves **uniformly** in a circle around the post. If the toy makes **five complete revolutions** every minute, **find the force *F*** exerted on it by the rope.

F=?

$$F = m\frac{4\pi^2 R}{T^2}$$

$$F = 25\frac{4\pi^2(5)}{(5/60)^2}$$



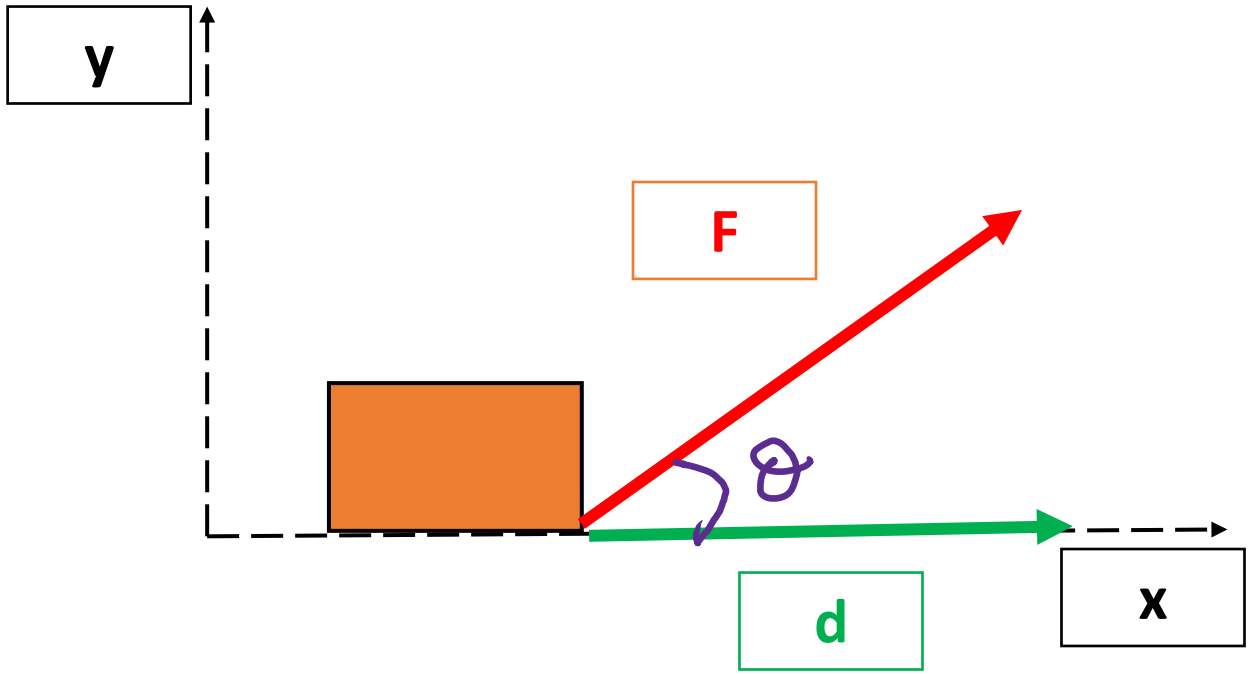
Work

W=?

$$W = \boldsymbol{F}.\boldsymbol{d}$$

$$W = F_x d$$

$$W = Fd\,Cos(\Theta)$$



Work

$\theta = 0$

$W = Fd$

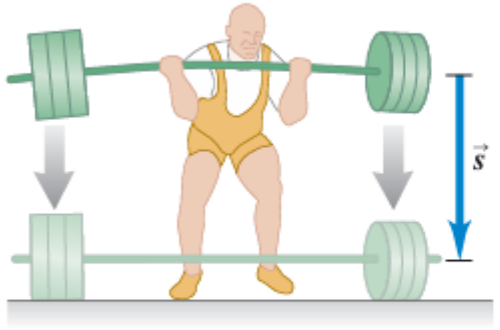
$\theta = 90$

$W = 0$

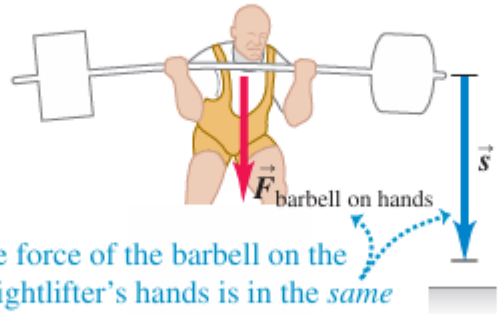
$\theta = 180$

$W = -Fd$

(a) A weightlifter lowers a barbell to the floor.

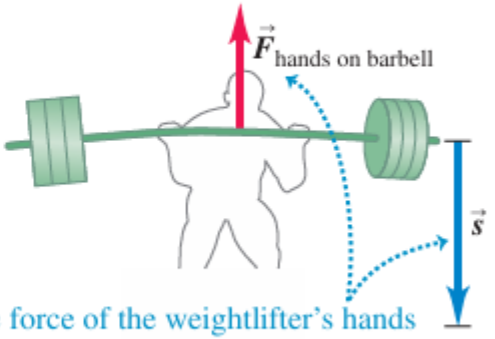


(b) The barbell does *positive* work on the weightlifter's hands.



The force of the barbell on the weightlifter's hands is in the *same* direction as the hands' displacement.

(c) The weightlifter's hands do *negative* work on the barbell.



The force of the weightlifter's hands on the barbell is *opposite* to the barbell's displacement.

d

x

Condition	Work	Example
$\cos\theta > 0$	Positive	Pushing a car forward
$\cos\theta < 0$	Negative	Braking a moving car
$\cos\theta = 0$	Zero	Carrying a load horizontally